



UTTAR PRADESH TEXTILE TECHNOLOGY INSTITUTE KANPUR
(Government Central Textile Institute Kanpur)

“Engineering Drawing ”

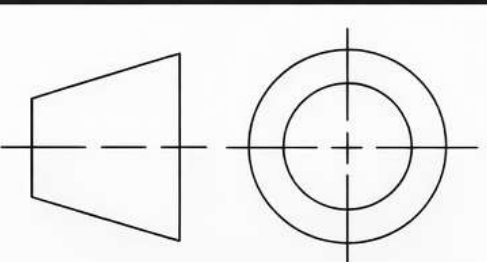
Dr. Vivek Mani Tripathi Assistant Professor, Department of Mechanical Engineering
UPTTI Kanpur
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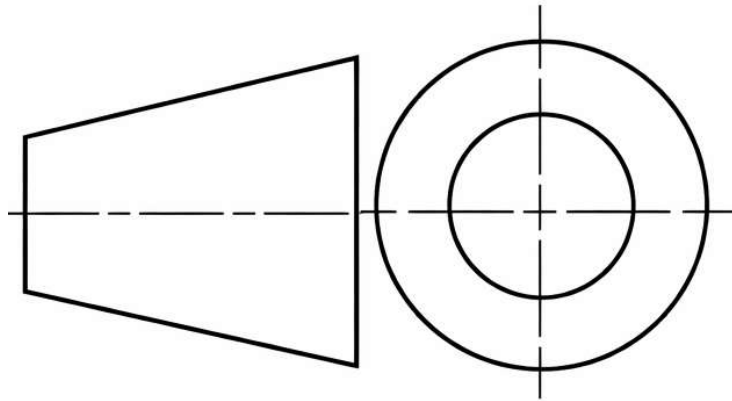
1. Lines & Dimensioning

Title Box

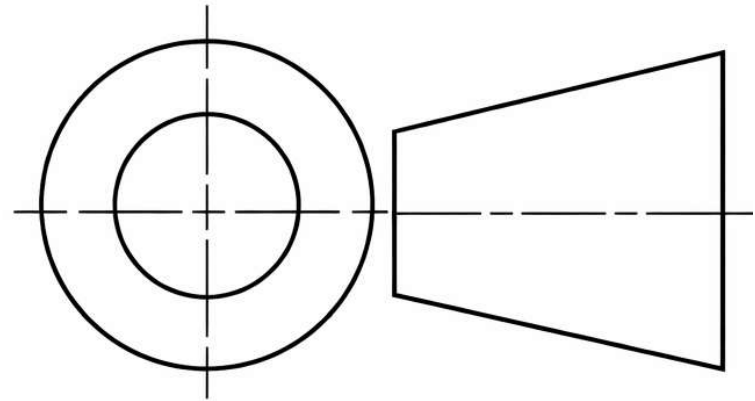
A **Title Box** is a **small information box** made on every engineering drawing.

50		120		
65	5	TITLE ORTHOGRAPHIC PROJECTIONS		
	5			
	5		NAME NAME OF STUDENT	
	5		CLASS BE I SEM MECH., Roll No.	
	5		COLLEGE NAME OF THE INSTITUTE	
	5			
	5	STARTED ON	SCALE 1:5	SHEET NO. 1 OF 1
	5	FINISHED ON	ALL DIMENSIONS ARE IN MILLIMETRES	
	5			
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5				
170				

Angle of projection



(a) Symbol for first angle projection

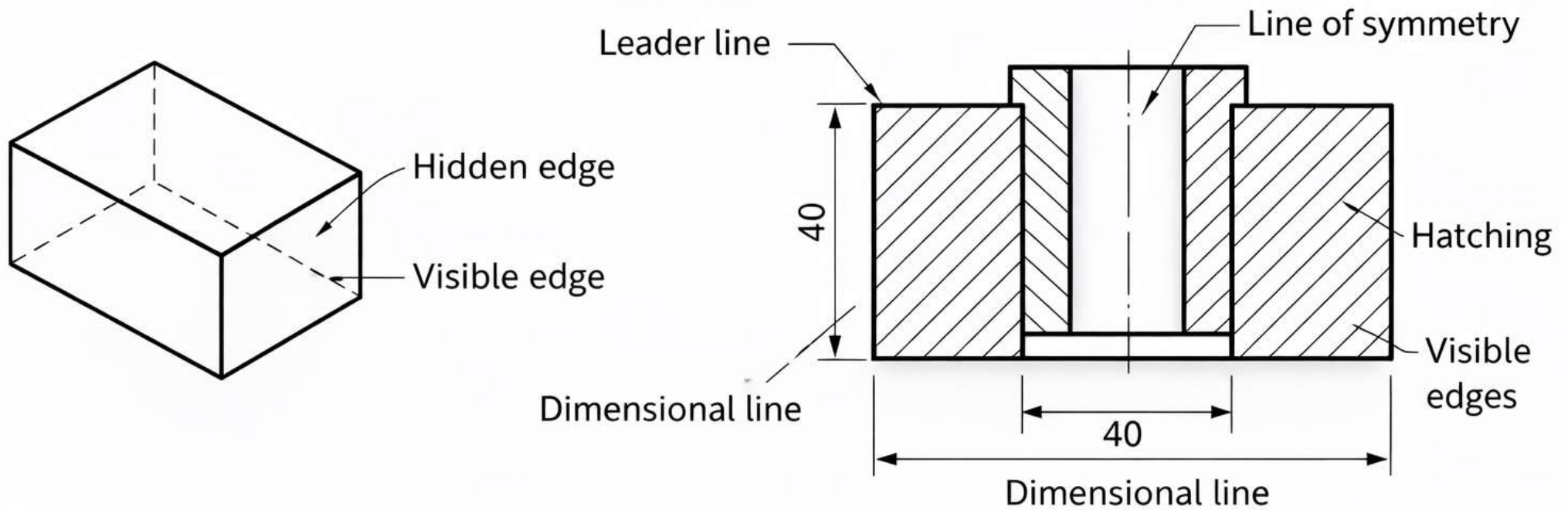


(b) Symbol for third angle projection

Lines

<i>S. No.</i>	<i>Representation</i>	<i>Description</i>
01	—————	Continuous line
02	- - - - -	Dashed line
03	- - - - - - - - -	Dashed space line
04	- - - - -	Long dashed dotted line
05	- - - - -	Long dashed double-dotted line
06		Long dashed triplicate-dotted line
07	-	Dotted line
08	- - - - -	Long dashed short dashed line
09	- - - - -	Long dashed double-short dashed line
10	-	Dashed dotted line
11	- - - - - - -	Double-dashed dotted line
12	- - - - -	Dashed double-dotted line
13	- - - - -	Double-dashed double-dotted line
14	- - - - -	Dashed triplicate-dotted line
15	- - - - -	Double-dashed triplicate-dotted line
16	- - - - -	Dashed triplicate triplicate-dotted line
15	- - - - -	Double-dashed triplicate-dotted line

Lines



Lines

<i>S. No.</i>	<i>Line description and representation</i>	<i>Applications</i>
1.	Continuous narrow lines 	(a) Imaginary lines of intersection (b) Dimension line (c) Extension line (d) Leader line with reference line (e) Hatching (f) Outlines of revolved section (g) Short centre lines (h) Root of screw threads (i) Dimension lines termination, arrowheads (j) Diagonals for indication of flat surfaces (k) Bending lines on blanks and processing parts (l) Framing of details (m) Indication of repetitive details (n) Interpretation lines of tapered features (o) Location of laminations (p) Projection lines (q) Grid lines

Lines

2.	Continuous narrow freehand line 	^a Preferably manually represented termination of partial or interrupted views, cuts and sections, if the limit is not a line of symmetry or a centre line
3.	Continuous narrow line with zigzags 	^a Preferably mechanically represented termination of partial or interrupted views, cuts and sections. If the limit is not a line of symmetry or a centre line
4.	Continuous wide line 	(a) Visible edges (b) Visible outlines (c) Crests of screw threads (d) Limit of length of full depth thread (e) Main representation in diagrams, maps, flow charts (f) System lines (structural metal engineering) (g) Parting lines of moulds in views (h) Lines of cuts and section arrows
5.	Dashed narrow line 	(a) Hidden edges (b) Hidden outlines
6.	Dashed wide line 	(a) Indication of permissible areas of surface treatment

Lines

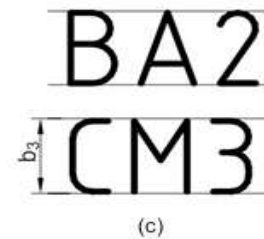
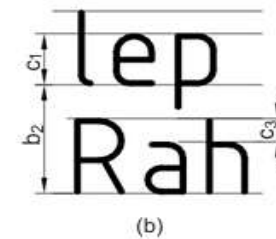
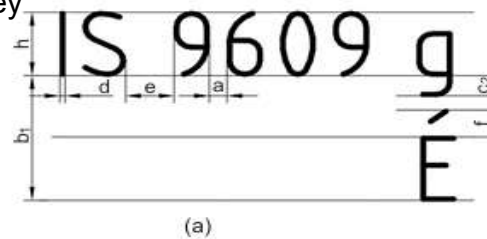
7.	Long-dashed dotted narrow line 	(a) Centre lines (b) Line of symmetry (c) Pitch circle of gears (d) Pitch circle of holes
8.	Long-dashed dotted wide line 	(a) Indication of (limited) required areas of surface treatment, eg. heat treatment (b) Indication of cutting planes
9.	Long-dashed double-dotted narrow line 	(a) Outlines of adjacent parts (b) Extreme positions of movable parts (c) Centroidal lines (d) Initial outlines prior to forming (e) Parts situated in front of a cutting plane (f) Outlines of alternative executions (g) Outlines of the finished part within blanks (h) Framing of particular fields/areas (i) Projected tolerance zone

LETTERING

Light guidelines drawn using sharp pencil ensures consistency in the size of the letter characters.

1. If the lettering consists of capitals, draw only the cap line and base line. If the lowercase letters are included as well, draw the waist line and the drop line.
2. The thickness of the line of the letter should be such as is obtained in one stroke of pencil. Exert a firm uniform pressure, but not so heavy as to cut grooves in the sheet.
3. The width-to-height ratio should be around 1:2 for all capital alphabets (except I and W) and 1:3 for all numerals (except 1).
4. Letters should be written in capitals. Lower-case alphabet should be used only when they are accepted in international usage for abbreviations.
5. The horizontal lines of the letter should be drawn from left to right and vertical or inclined lines from top to bottom. Alphabet and numerals should neither touch each other nor the lines.
6. Letters and numerals should neither touch each other nor the lines.
7. Letters should be so spaced that the area between letters appears equal. It is not necessary to keep clearances between adjacent letters equal. For example, letters LA, TV or Tr.
8. Words should be spaced one letter apart.
9. Letters should be so written that they appear upright from the bottom edge, except when they are used for dimensioning. For dimensioning, they may appear upright from the bottom edge for the right hand side or the corner in between

Characteristic	Ratio	Dimensions (mm)							
Height of capital lettering, h	$\left(\frac{14}{14}\right)h$	1.8	2.5	3.5	5	7	10	14	20
Height of lower-case lettering, c_1	$\left(\frac{10}{14}\right)h$	1.3	1.8	2.5	3.5	5	7	10	14
Tails of lower case letters, c_2	$\left(\frac{4}{14}\right)h$	0.52	0.72	1	1.4	2	2.8	4	5.6
Stem of lower case letters, c_3	$\left(\frac{4}{14}\right)h$	0.52	0.72	1	1.4	2	2.8	4	5.6
Area of diacritical marks (upper case letters), f	$\left(\frac{5}{14}\right)h$	0.65	0.9	1.25	1.75	2.5	3.5	5	7
Spacing between characters, a	$\left(\frac{2}{14}\right)h$	0.26	0.36	0.5	0.7	1	1.4	2	2.8
Minimum spacing of baselines ¹⁾ , b_1	$\left(\frac{25}{14}\right)h$	3.25	4.5	6.25	8.75	12.5	17.5	25	35
Minimum spacing of baselines ²⁾ , b_2	$\left(\frac{21}{14}\right)h$	2.73	3.78	5.25	7.35	10.5	14.7	21	29.4
Minimum spacing of baselines ³⁾ , b_3	$\left(\frac{17}{14}\right)h$	2.21	3.06	4.25	5.95	8.5	11.9	17	23.8
Minimum spacing between words, e	$\left(\frac{6}{14}\right)h$	0.78	1.08	1.5	2.1	3	4.2	6	8.4
Line width, d	$\left(\frac{1}{14}\right)h$	0.13	0.18	0.25	0.35	0.5	0.7	1	1.4



LETTERING

A B C D E F G H I J K L M N

O P Q R S T U V W X Y Z

a b c d e f g h i j k l m n o p

q r s t u v w x y z

0 1 2 3 4 5 6 7 8 9 I V X

0 1 2 3 4 5 6 7 8 9 I V X

(a)

A B C D E F G H I J K L M N

O P Q R S T U V W X Y Z

a b c d e f g h i j k l m n o p

q r s t u v w x y z

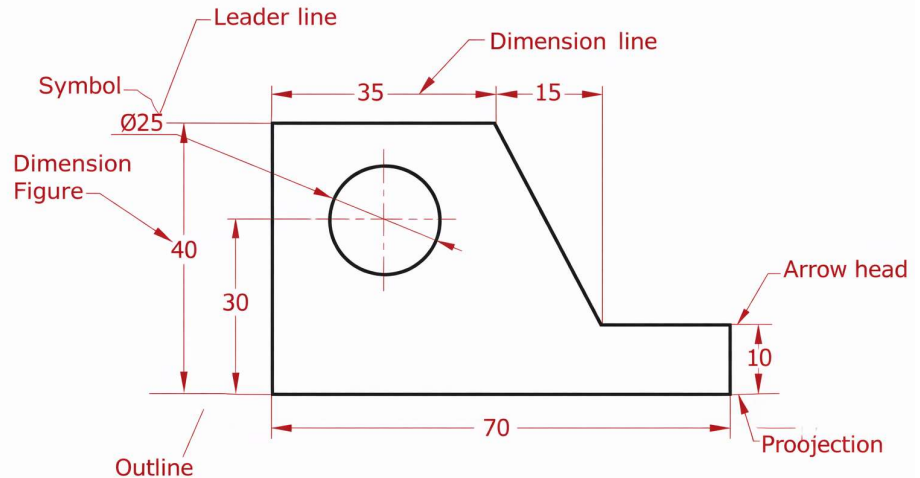
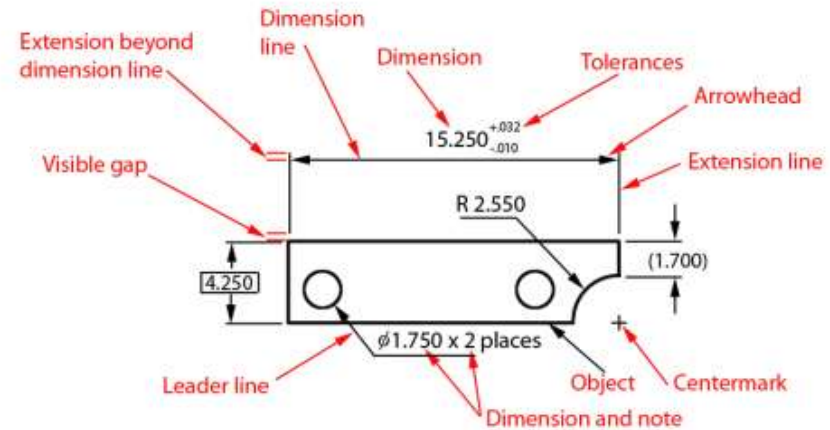
0 1 2 3 4 5 6 7 8 9 I V X

0 1 2 3 4 5 6 7 8 9 I V X

(b)

DIMENSIONING

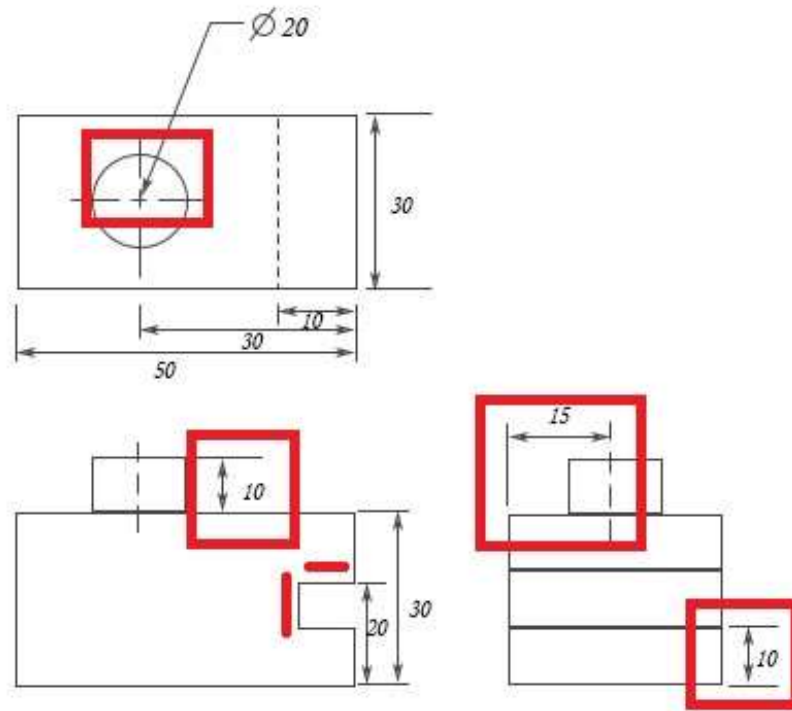
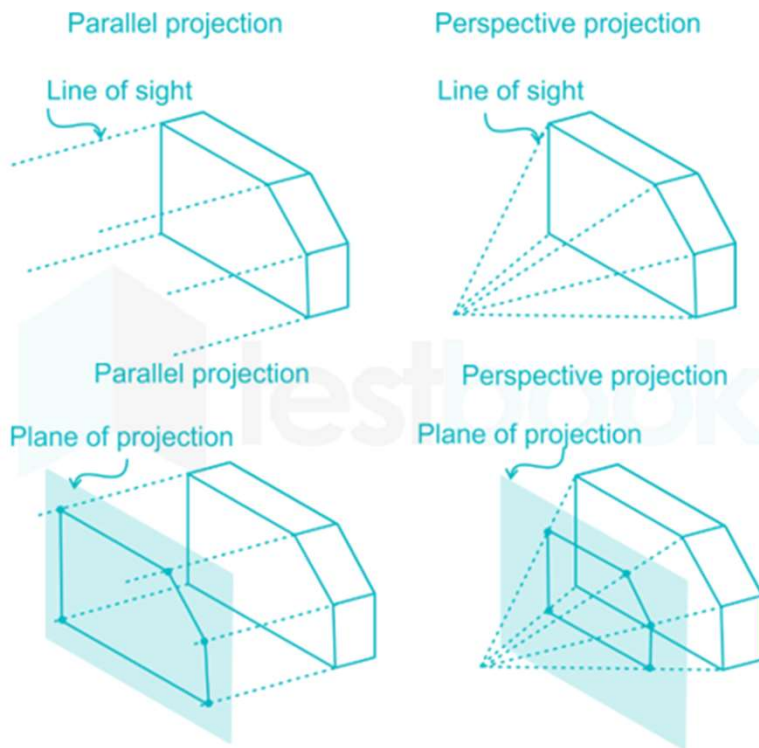
- **Dimension Line** → Shows the size (with number)
- **Arrowhead** → Shows where measurement starts and ends
- **Projection Line (Extension Line)** → Extends from object to dimension line
- **Leader Line** → Points to special notes or holes
- **Centre Line** → Shows middle of a circle/hole
- **Outline** → Outer boundary of object



DIMENSIONING

They must be **parallel to each other**

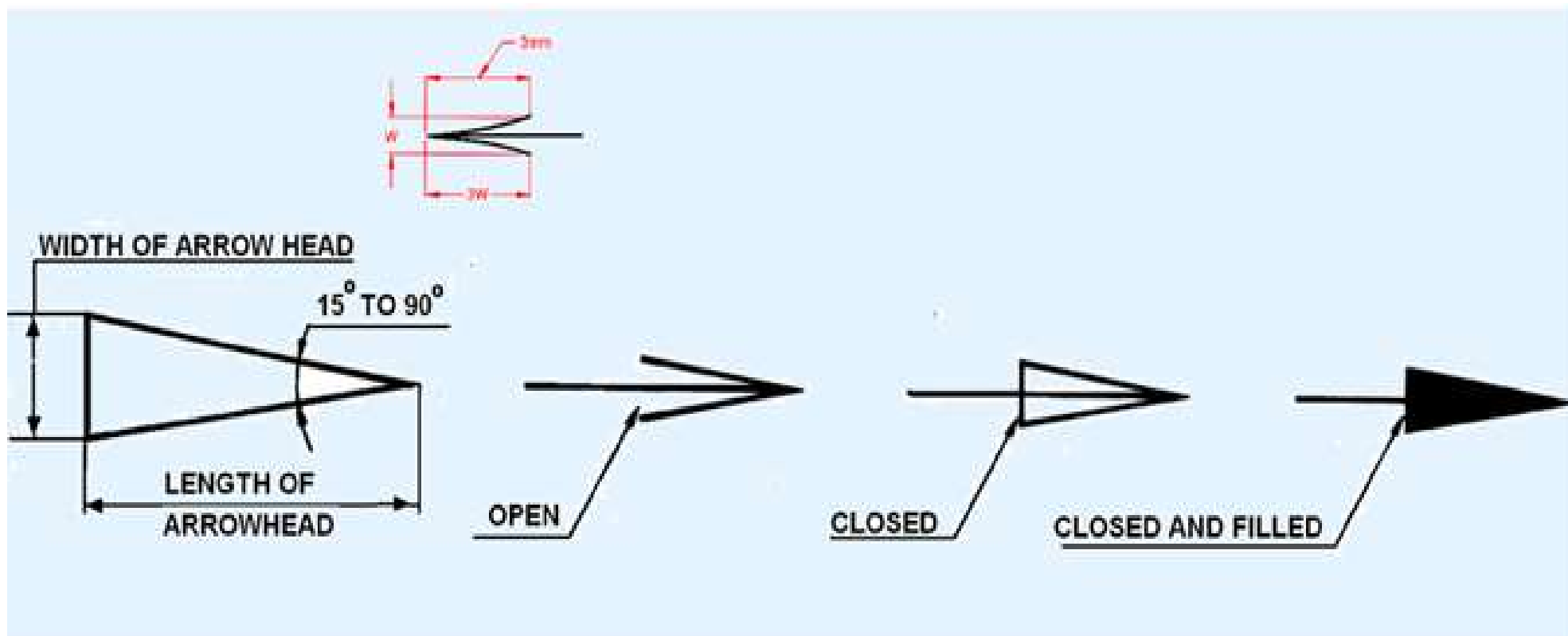
✓ Used only when needed



DIMENSIONING

Types of Arrowheads

- 1 Open 90°
 - 2 Open 30°
 - 3 Closed blank
 - 4 Closed filled (most common)
- 👉 Closed filled is mostly used in engineering drawings.



2. Orthographic Projection

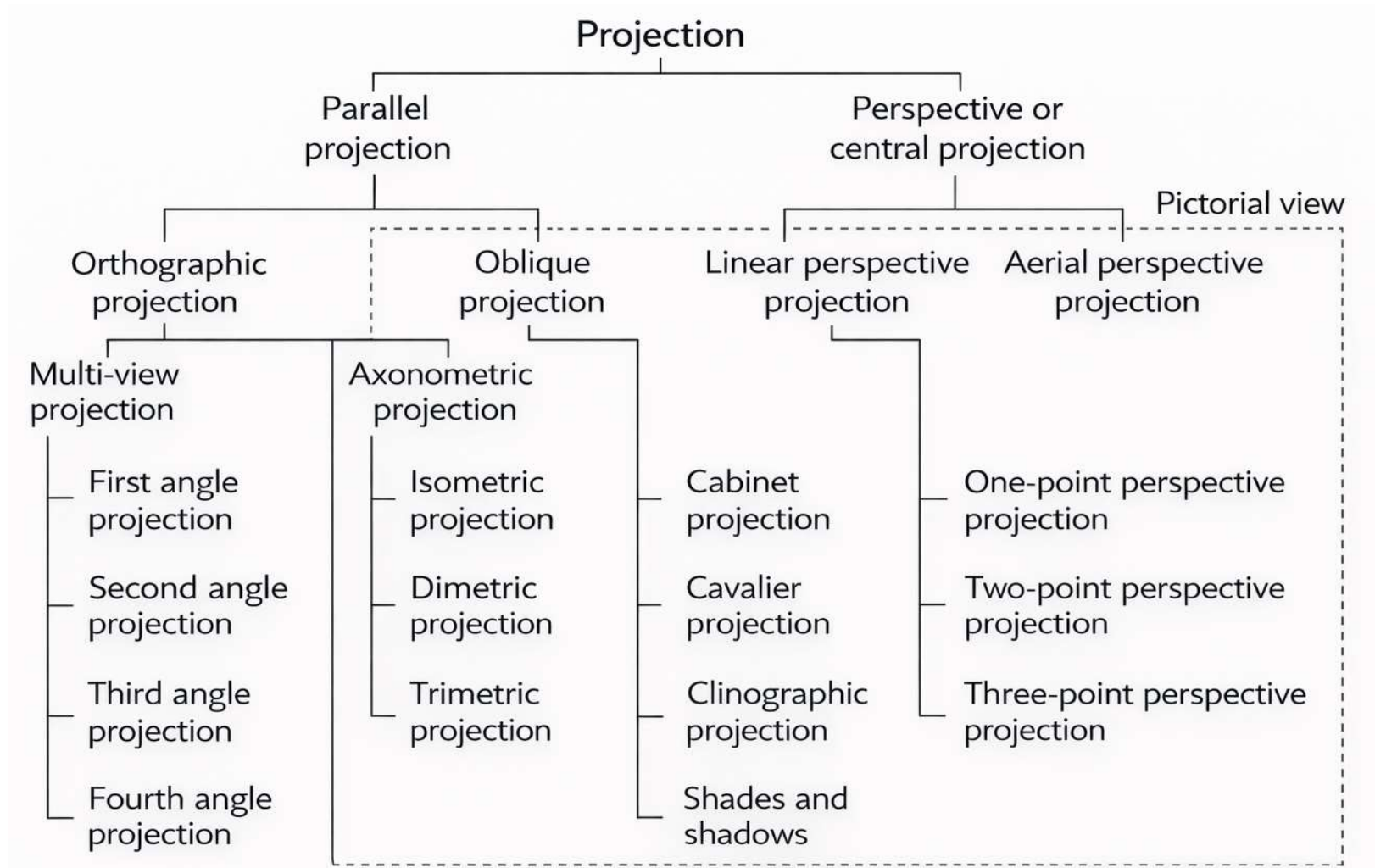
Projection

A projection is a picture or drawing of an object made on a flat surface (plane). In engineering, almost all drawings are made using the principle of projection.

This method helps to show the exact shape, size, and details of an object. With projection, both the outer features and internal parts of an object can be clearly represented in a drawing.

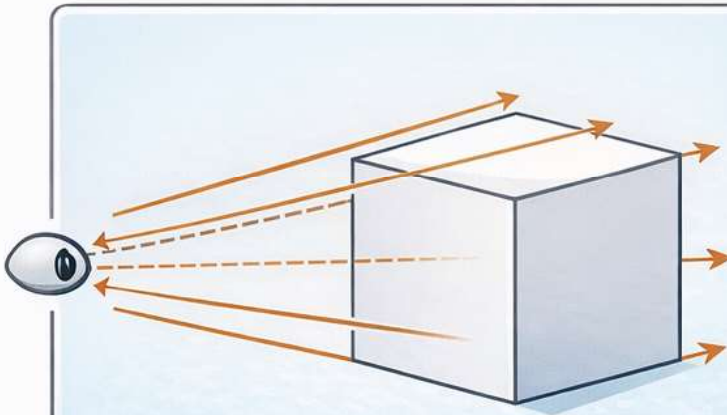
Projections can be classified based on **two factors**:

- 1.The **direction of the lines of sight** used to view the object.
- 2.The **position of the plane** on which the drawing or image is formed



Parallel Projection vs Perspective Projection

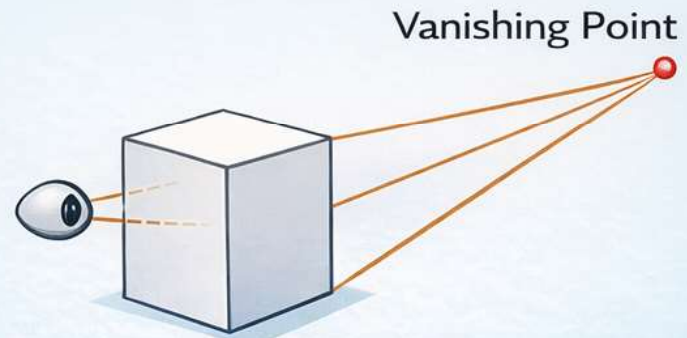
Parallel Projection



- Lines do not meet.
- Object size stays the same.

VS

Perspective Projection



- Lines meet at a point.
- Objects look smaller further away.

Feature	Parallel Projection	Perspective Projection
Projection lines	Lines of sight are parallel to each other	Lines of sight meet at a single point (called the center of projection)
Appearance	Object size remains same even if it is far away	Objects appear smaller when they are far away
Realism	Less realistic but more accurate for measurements	More realistic and natural view
Use	Used in engineering drawings and technical drawings	Used in art, architecture, and visual presentations
Example	Orthographic projection, Isometric projection	One-point, Two-point, Three-point perspective

PICTORIAL VIEW AND MULTI-VIEW

When we draw a **3D object on a 2D paper**, we use different projection methods to show its shape.

The four drawings are:

- (a) **Perspective projection**
- (b) **Axonometric projection**
- (c) **Oblique projection**
- (d) **Multi-view projection**

Each method shows the object in a different way.

(a) Perspective Projection

In **perspective projection**, the object looks **like we see it with our eyes**.

Simple points:

- Near parts look **bigger**
- Far parts look **smaller**
- It looks **very realistic**

Example: When you see a road, it looks **narrower far away**.

So this method is used for:

- Architecture
- Photography
- Realistic drawings

(b) Axonometric Projection

In **axonometric projection**, the object is drawn so that:

- All parts remain the same size**

- Near and far parts **do not change size**

So the object looks **slightly realistic but still accurate**.

Common example:

- Isometric drawing**

This is widely used in **engineering drawings**.

PICTORIAL VIEW AND MULTI-VIEW

(c) Oblique Projection

In **oblique projection**:

- The **front face is drawn in the true shape and size**
- The **depth is drawn at an angle**

Advantages:

- Easy to draw
- Curves and circles in the front face are easy to show

But the **side faces look slightly distorted**.

(d) Multi-View Projection

In **multi-view projection**, the object is shown using **different views**:

Example:

- Front view
- Top view
- Side view

Instead of one picture, **separate views are drawn**.

This method is **most accurate in engineering**.

Type

Perspective

Axonometric

Oblique

Multi-view

Appearance

Realistic

Equal size

True front face

Separate views

Use

Architecture, design

Engineering drawings

Simple sketches

Manufacturing drawings

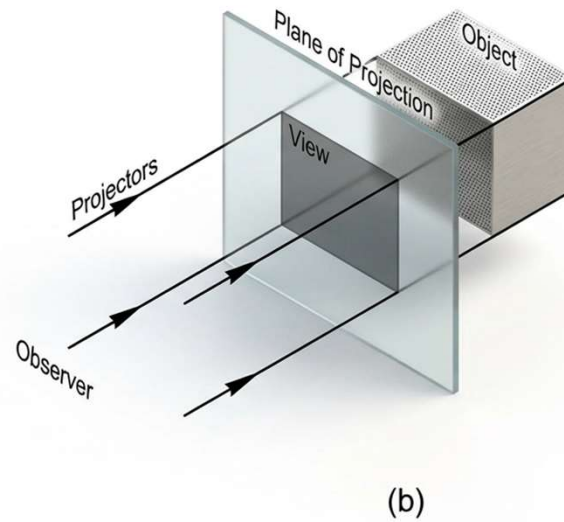
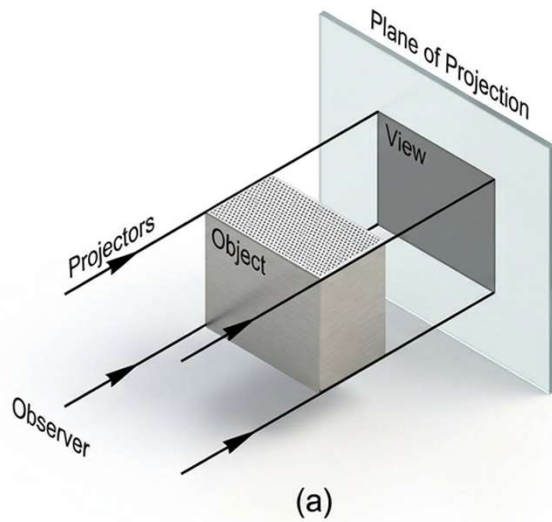
ORTHOGRAPHIC PROJECTION

The word **orthographic** comes from the Greek word “**orthos**”, which means **perpendicular**. In orthographic projection, we imagine that the **observer is very far away from the object**.

Because the observer is far away:

- The **lines of sight (rays from the eye)** become **parallel to each other**.
- These lines are also **perpendicular to the plane of projection**.

These lines are called **projectors**.



Orthographic projection is a type of parallel projection in which the projectors are parallel to each other and perpendicular to the projection plane.

What Orthographic Projection Shows

Using orthographic projection, we can draw objects in two ways:

Pictorial View

- Shows **all three dimensions (length, width, height)** in **one drawing**.

Example:

Isometric drawing

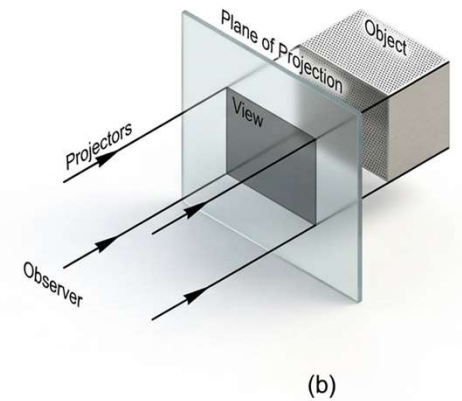
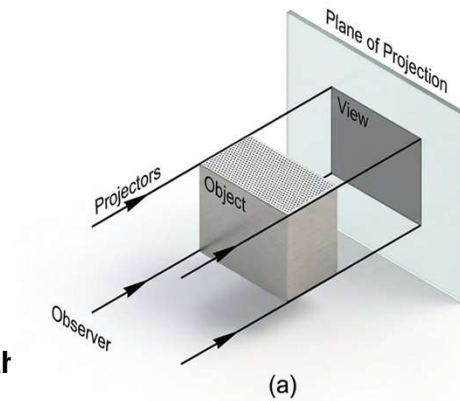
Multi-View Drawing

- The object is shown using **separate views**.
- Each view shows **only two dimensions**.

Common views are:

- Front view**
- Top view**
- Side view**

These views together help to understand the **complete shape of the object**.



Orthographic projection is a method of drawing where **parallel projectors perpendicular to the projection plane** are used to show the **exact shape and size of an object**.

MULTI-VIEW DRAWING

A **multi-view drawing** is used to show the **complete shape of a 3D object**.

Since a single drawing cannot show all three dimensions clearly, we draw **two or more views** of the same object.

Each view shows **only two dimensions** of the object.

For example:

- **Front view** → shows **height and width**
- **Top view** → shows **length and width**
- **Side view** → shows **height and length**

Because each view shows only part of the information, **all the views must be studied together** to understand the complete object.

Orthographic Views

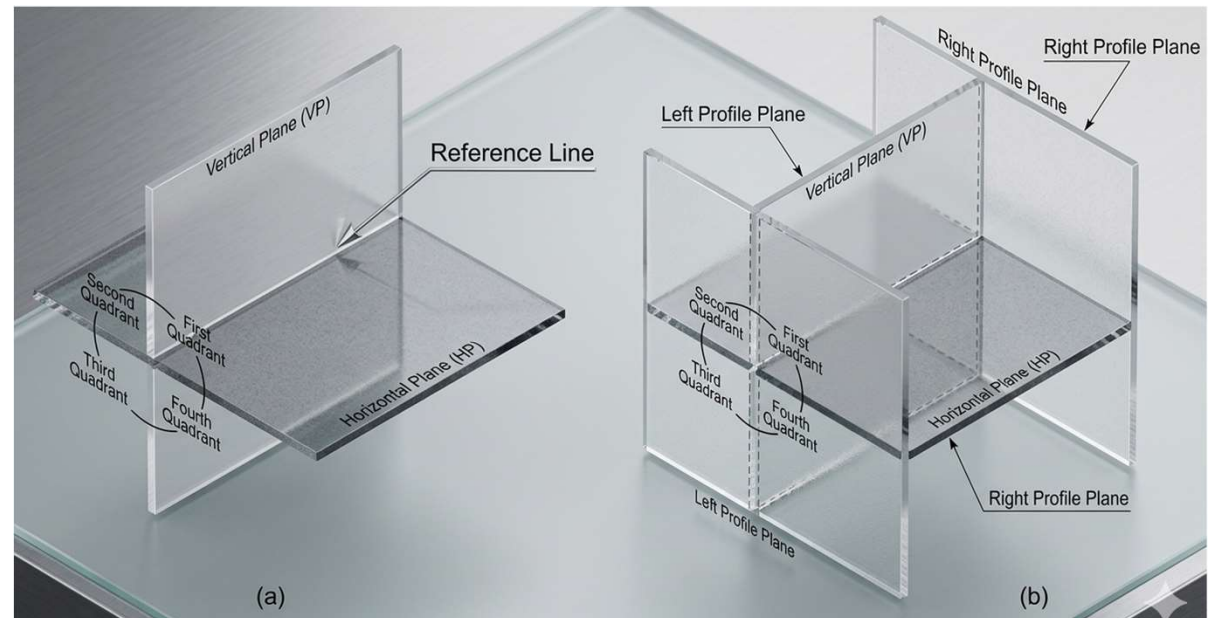
These views are called **orthographic projections**.

They are drawn on two important reference planes:

1. **Vertical Plane (VP)**
2. **Horizontal Plane (HP)**

Relationship Between VP and HP

- The **Vertical Plane** stands **upright** (like a wall).
- The **Horizontal Plane** lies **flat** (like the floor).
- These two planes are **perpendicular (90°)** to each other.



MULTI-VIEW DRAWING

Quadrants (Angles)

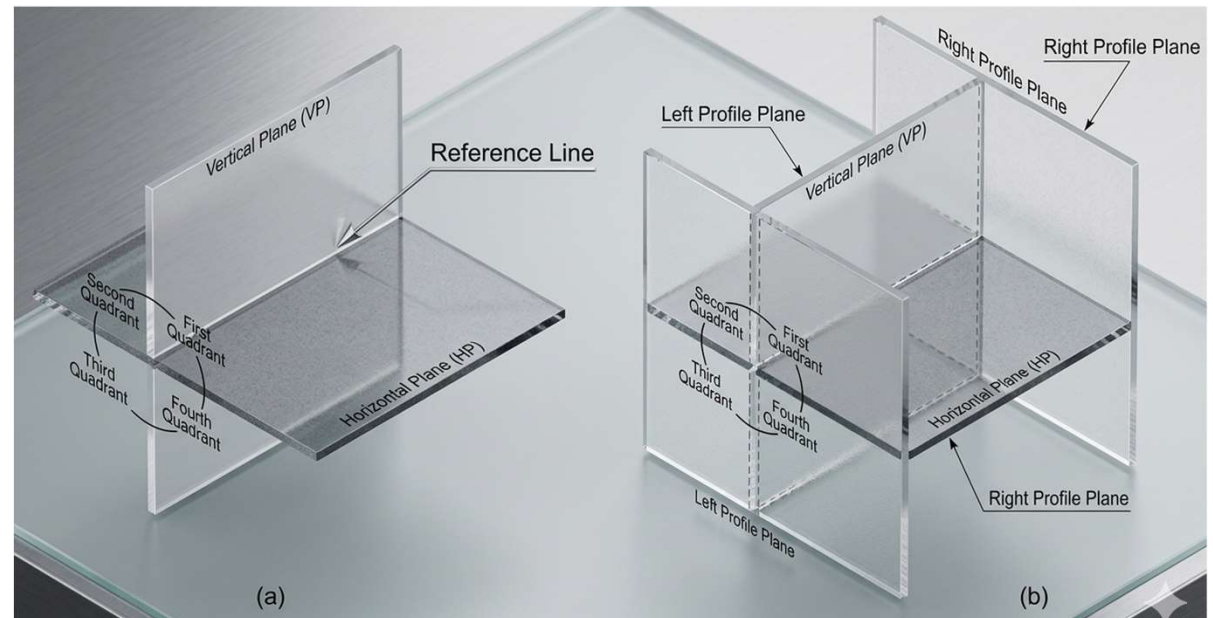
When these two planes intersect:

- The space is divided into **four parts**.
- These parts are called **quadrants** or **angles**.

They are:

- 1.**First Quadrant**
- 2.**Second Quadrant**
- 3.**Third Quadrant**
- 4.**Fourth Quadrant**

These quadrants help in **deciding where the object is placed and how the views are drawn**.



MULTI-VIEW DRAWING

First Angle

- Object is **above the Horizontal Plane (H.P.)**
- Object is **in front of the Vertical Plane (V.P.)**

This region is called the **First Angle**.

👉 Used in **First Angle Projection**.

Third Angle

- Object is **below the Horizontal Plane (H.P.)**
- Object is **behind the Vertical Plane (V.P.)**

This region is called the **Third Angle**.

👉 Used in **Third Angle Projection**.

Second Angle

- Object is **above the Horizontal Plane (H.P.)**
- Object is **behind the Vertical Plane (V.P.)**

This region is called the **Second Angle**.

👉 Used in **Second Angle Projection**.

Fourth Angle

- Object is **below the Horizontal Plane (H.P.)**
- Object is **in front of the Vertical Plane (V.P.)**

This region is called the **Fourth Angle**.

👉 Used in **Fourth Angle Projection**

Angle

First Angle

Second Angle

Third Angle

Fourth Angle

Position of Object

Above H.P. and in front of V.P.

Above H.P. and behind V.P.

Below H.P. and behind V.P.

Below H.P. and in front of V.P.

FIRST ANGLE PROJECTION

In **first angle projection**, the object is placed:

- Above the Horizontal Plane (H.P.)
- In front of the Vertical Plane (V.P.)

Front View

The observer looks at the object **from the front side**.

- The rays of sight are **parallel** and **perpendicular to the Vertical Plane (V.P.)**.
- The image formed on the V.P. is called the **Front View**.

Front view shows:

- Length**
- Height**

Front view **does not show width**.

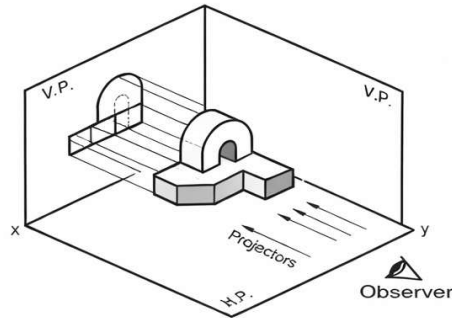
Now the observer looks at the object **from the top**.

- The rays of sight are **parallel** and **perpendicular to the Horizontal Plane (H.P.)**.
- The image formed on the H.P. is called the **Top View**.

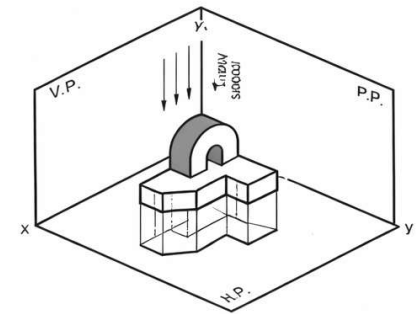
Top view shows:

- Length**
- Width**

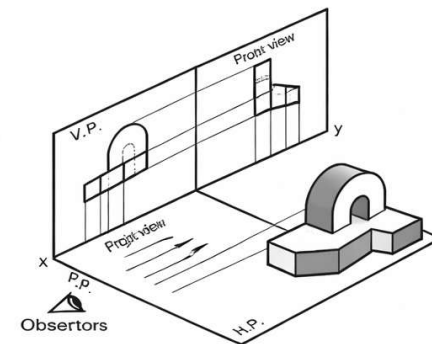
Top view **does not show height**.



(a) Front view on V.P. (Corrected Observer)



(b) Top view on H.P.



(c) Left-hand side view on P.P. (Corrected Observer & View)

FIRST ANGLE PROJECTION

Side View

Next the observer looks at the object **from the left side**.

- The rays of sight are **parallel** and **perpendicular to the Profile Plane (P.P.)**.

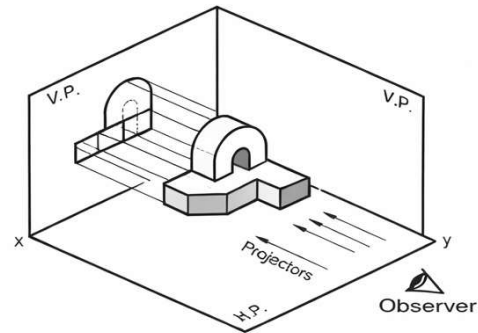
- The image formed on the P.P. is called the **Side View**.

Side view shows:

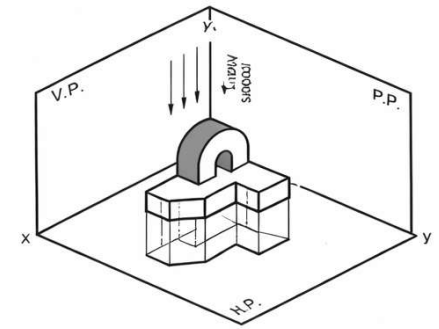
- Width**

- Height**

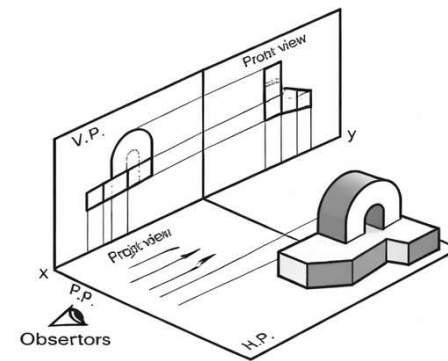
Side view **does not show length**.



(a) Front view on V.P. (Corrected Observer)

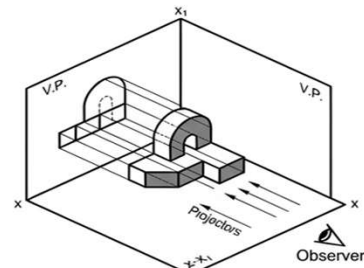


(b) Top view on H.P.

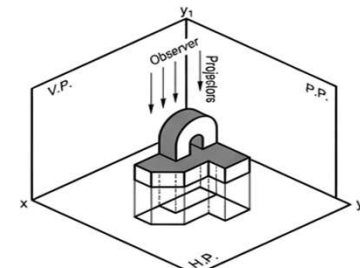


(c) Left-hand side view on P.P. (Corrected Observer & View)

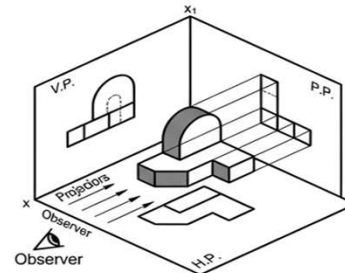
FIRST ANGLE PROJECTION



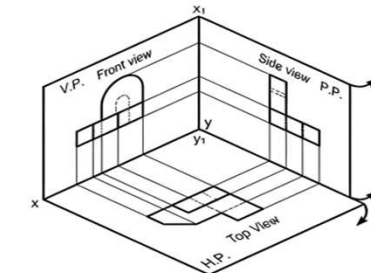
(a) Front view on V.P. (Corrected Observer)



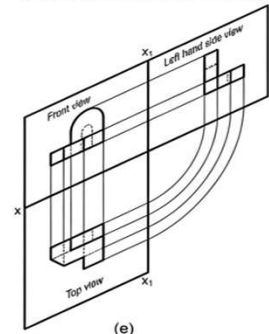
(b) Top view on H.P.



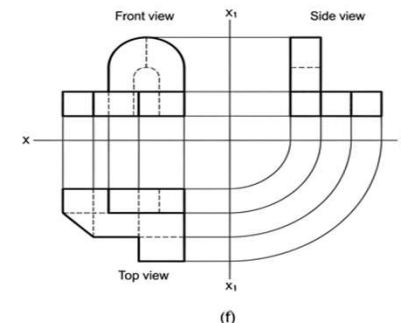
(c) Left-hand side view on P.P. (Corrected Observer & View)



(d)



(e)



(f)

View

Front View

Top View

Side View

Shows

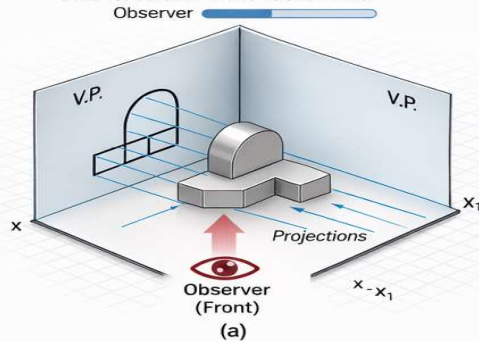
Length + Height

Length + Width

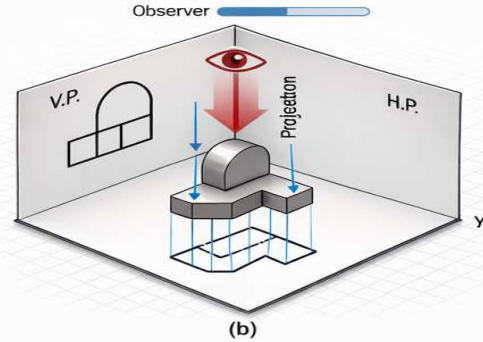
Width + Height

FIRST ANGLE PROJECTION

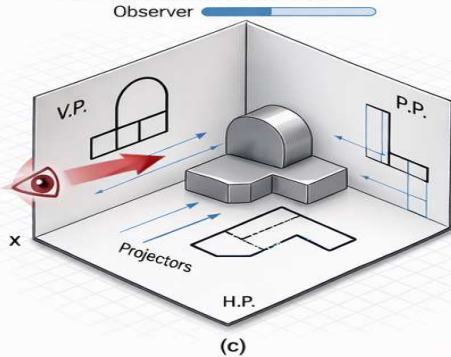
STEP 1: FRONT VIEW ISOLATION



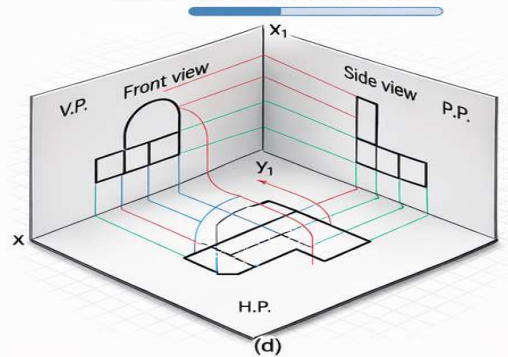
STEP 2: TOP VIEW ISOLATION



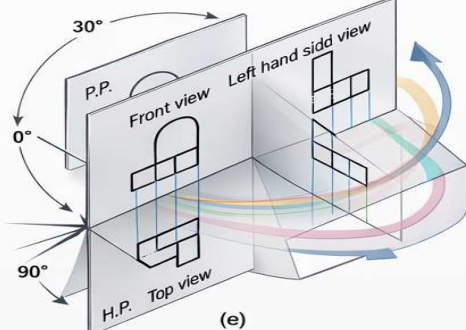
STEP 3: SIDE VIEW ISOLATION



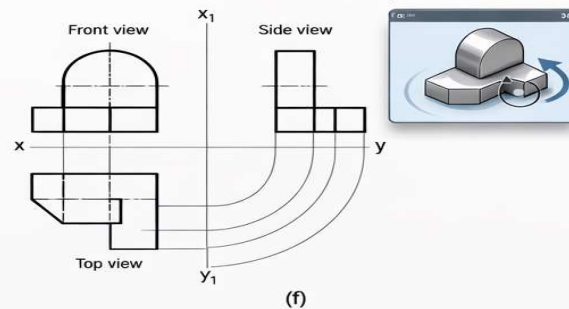
STEP 4: INTER-VIEW CORRELATION



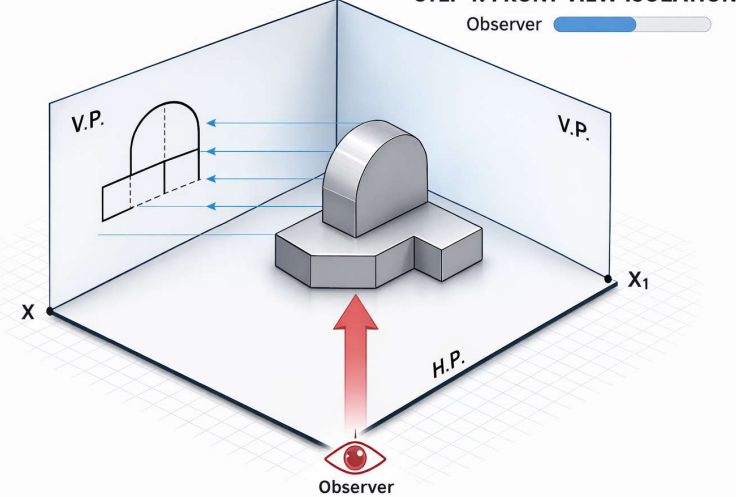
STEP 5: SIMULATED UNROLLING



FINAL ORTHOGRAPHIC REPRESENTATION (3D CAD INTEGRATED)



STEP 1: FRONT VIEW ISOLATION



THIRD ANGLE PROJECTION

In **third angle projection**, the object is placed:

- **Behind the Vertical Plane (V.P.)**
- **Below the Horizontal Plane (H.P.)**

This position is called the **third angle (third quadrant)**.

Front View

The observer looks at the object **from the front side**.

- Rays of sight are **parallel to each other**.
 - They are **perpendicular to the Vertical Plane (V.P.)**.
- The image formed on **V.P.** is called the **Front View**.

Front view shows:

- **Length**
- **Height**

Front view **does NOT** show width.

Now the observer looks at the object **from the top**.

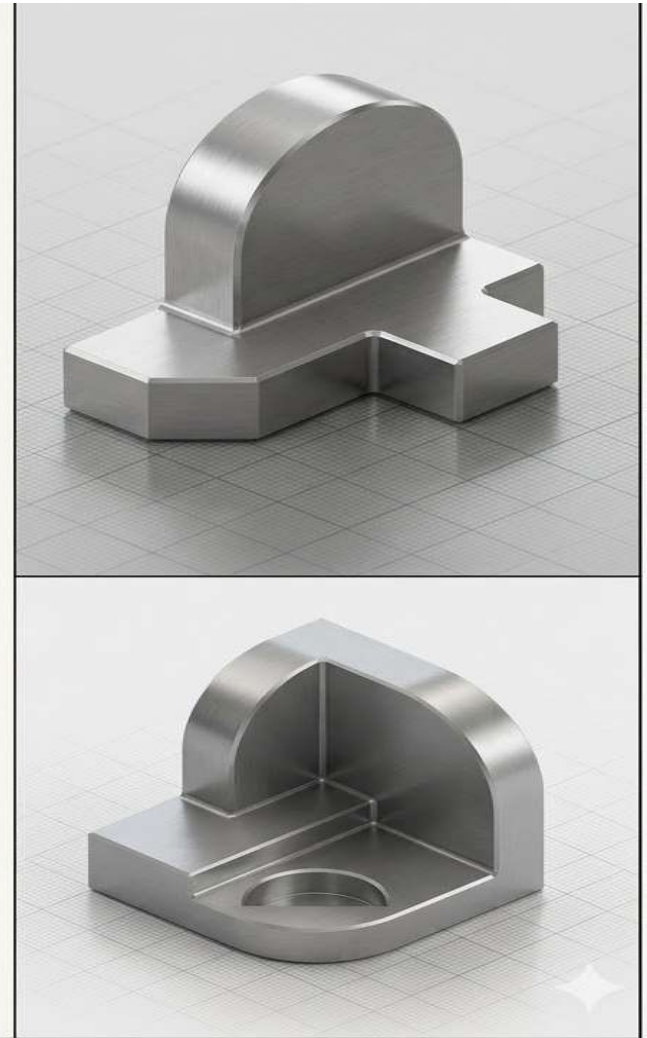
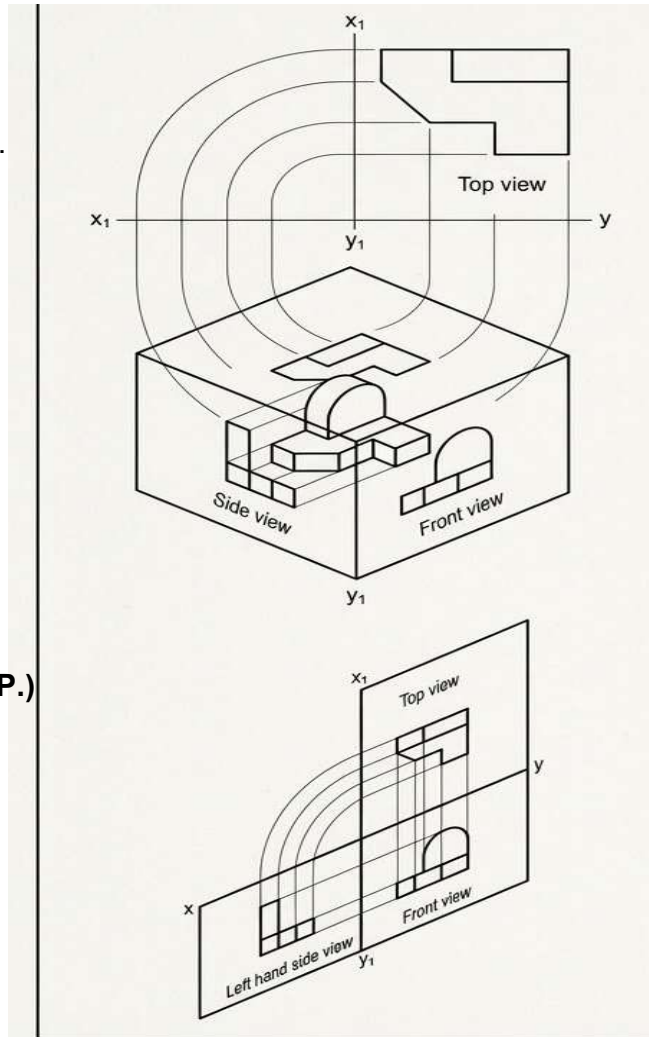
- Rays are **parallel**
- Rays are **perpendicular to the Horizontal Plane (H.P.)**

The image formed on **H.P.** is called the **Top View**.

Top view shows:

- **Length**
- **Width**

Top view **does NOT** show height



THIRD ANGLE PROJECTION

Side View

Now the observer looks at the object **from the left side**.

- Rays are **parallel**
- Rays are **perpendicular to the Profile Plane (P.P.)**

The image formed on **P.P.** is called the **Side View**.

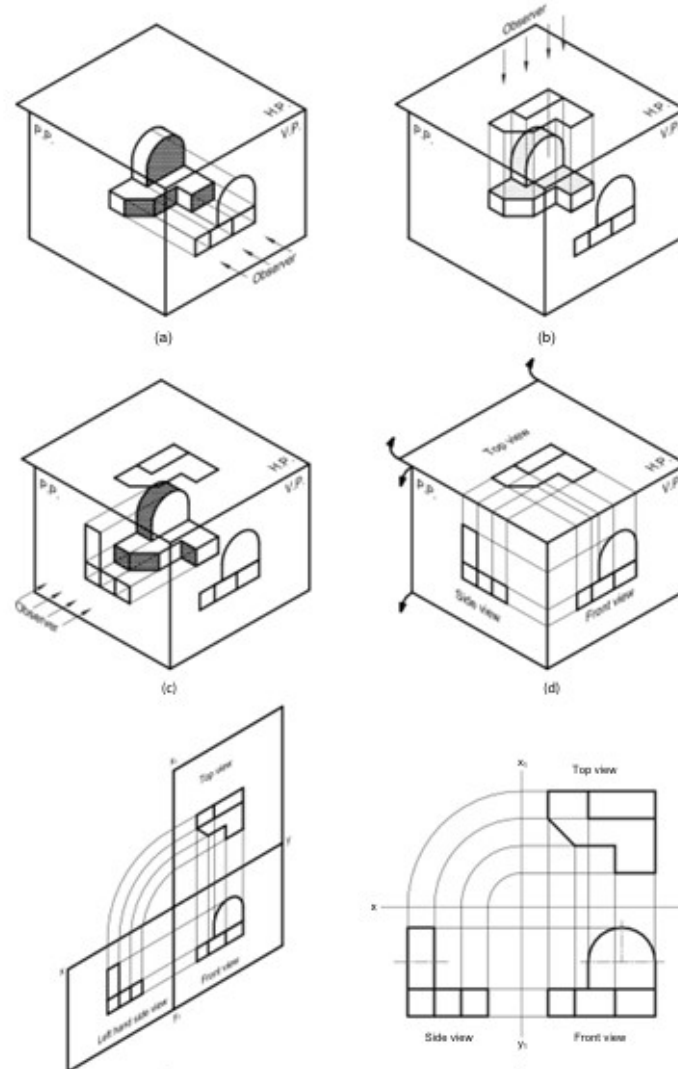
Side view shows:

- **Width**
- **Height**

Side view **does NOT** show length

In **Third Angle Projection**:

- **Top view** is drawn **above** the front view
- **Right side view** is drawn on the **right side**



Comparison of First and Third Angle Projection

Feature	First Angle Projection	Third Angle Projection
Quadrant	The object is placed in the First Quadrant .	The object is placed in the Third Quadrant .
Sequence	Observer Object Plane	Observer Plane Object
Plane Type	The projection plane is assumed to be opaque (non-transparent).	The projection plane is assumed to be transparent .
Positioning	The object lies between the observer and the plane.	The plane lies between the observer and the object.
Top View	Placed below the Front View.	Placed above the Front View.
Side View	The Left-hand side view is drawn on the right of the front view.	The Left-hand side view is drawn on the left of the front view.
Global Standard	Widely used in India, UK, and Europe (ISO).	Widely used in the USA, Canada, and Australia (ANSI).

SYMBOLS

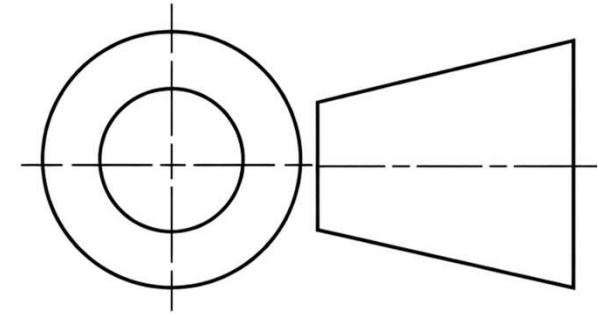
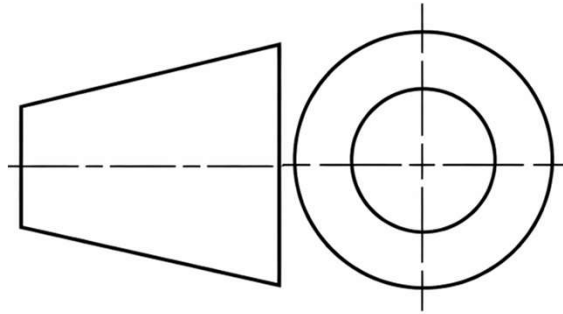
When we draw **multi-view drawings** (front view, top view, side view), the views should **not overlap each other**.

If views overlap, the drawing becomes **confusing and difficult to understand**.

To avoid this problem, internationally **only two projection methods are used**:

1.First Angle Projection

2.Third Angle Projection



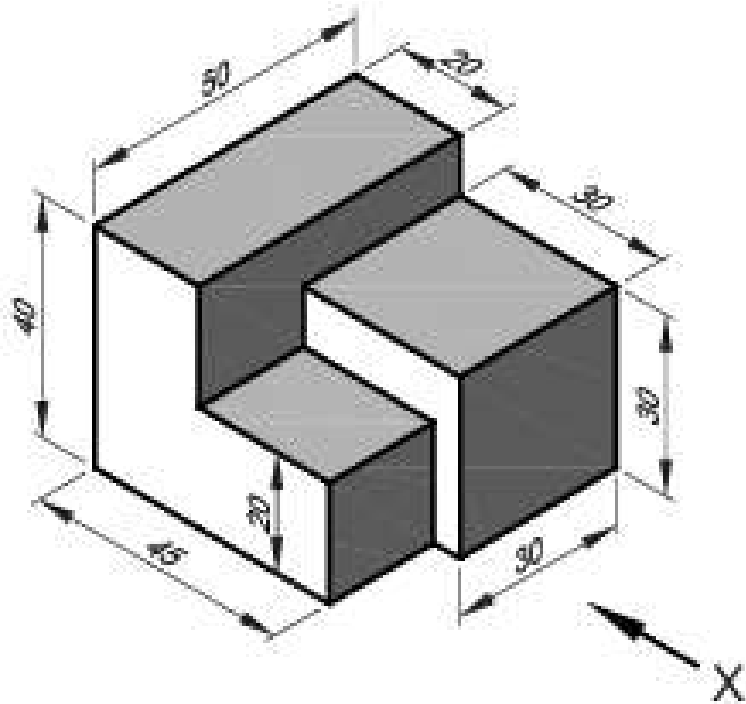
The symbol is made using the **views of a frustum of a cone**.

A **frustum of a cone** means a **cone whose top part is cut off**.

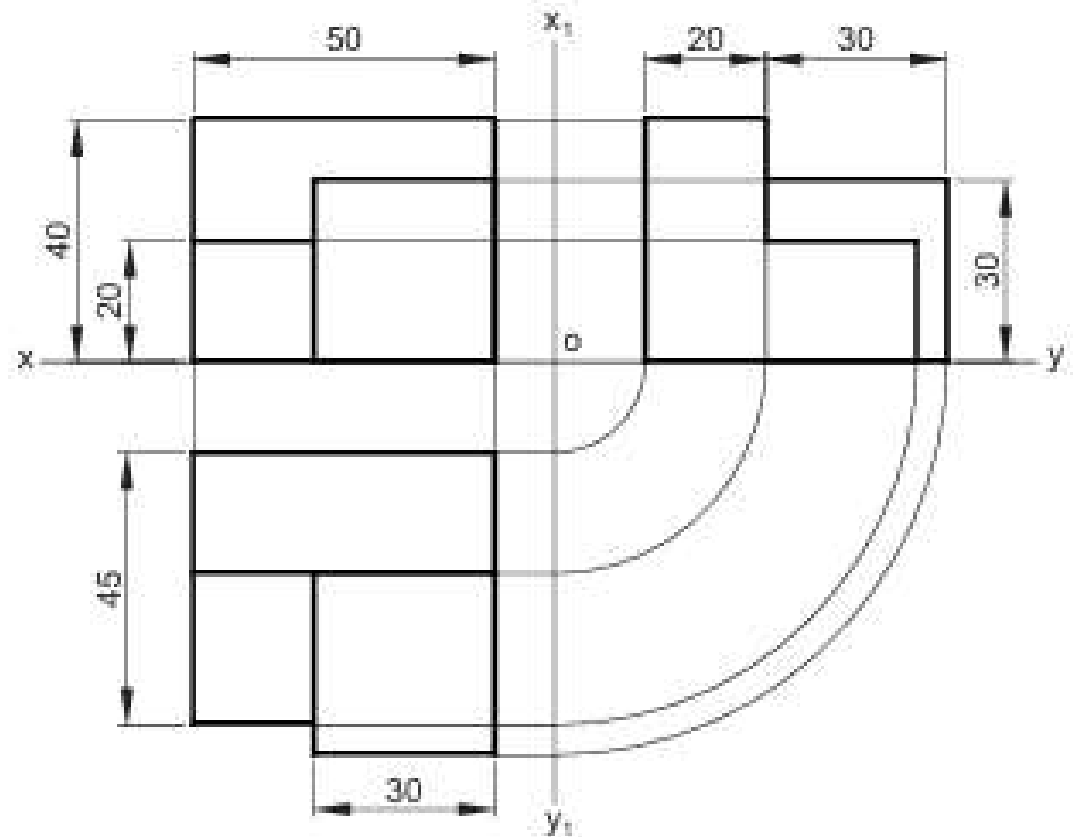
In this symbol:

- The **diameters of the cone ends are in the ratio 1 : 2**
- The **length of the cone equals the diameter of the bigger end**

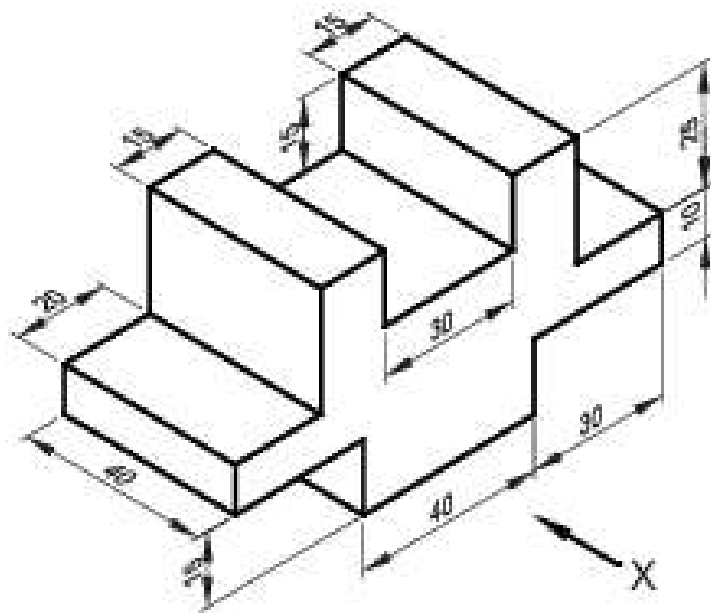
Questions



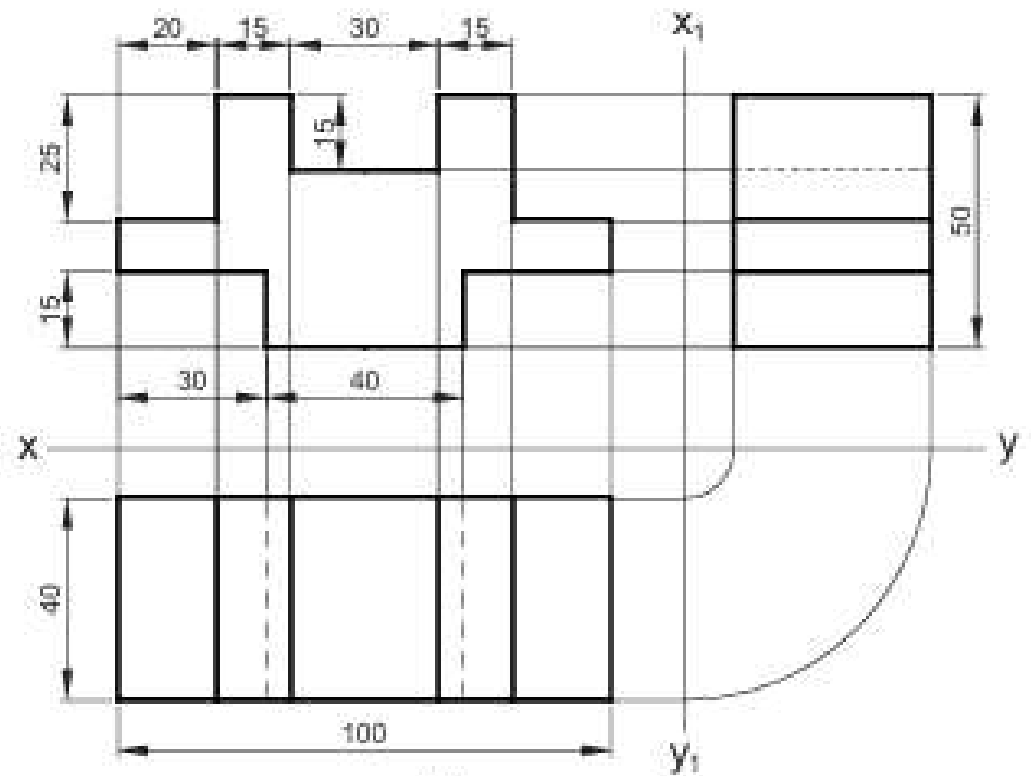
(a)



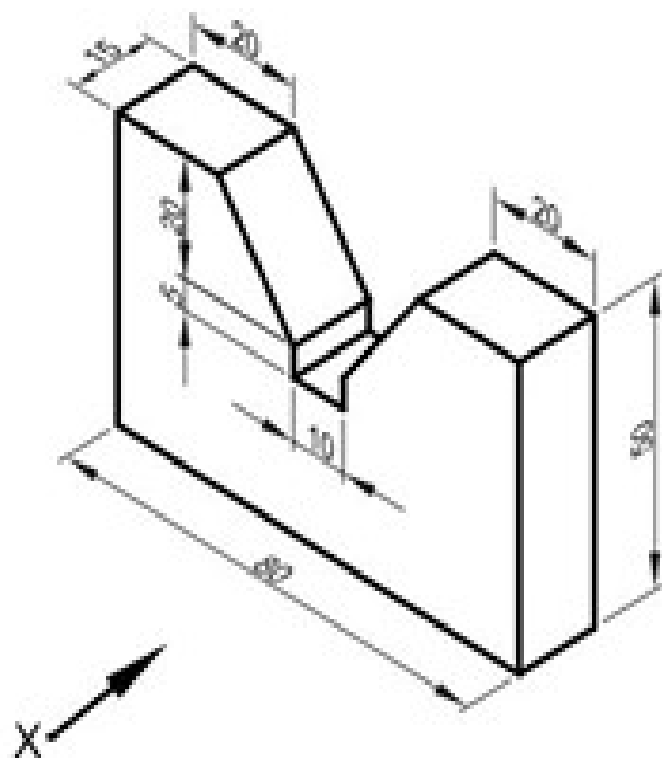
(b)



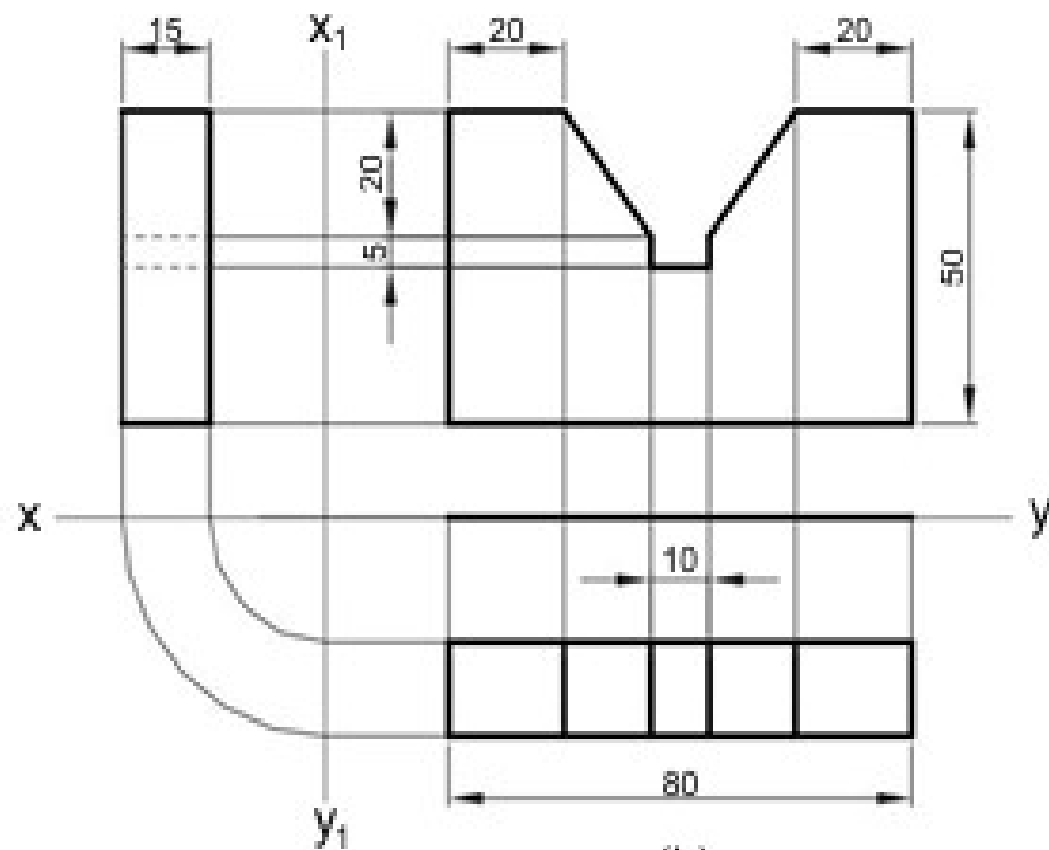
(a)



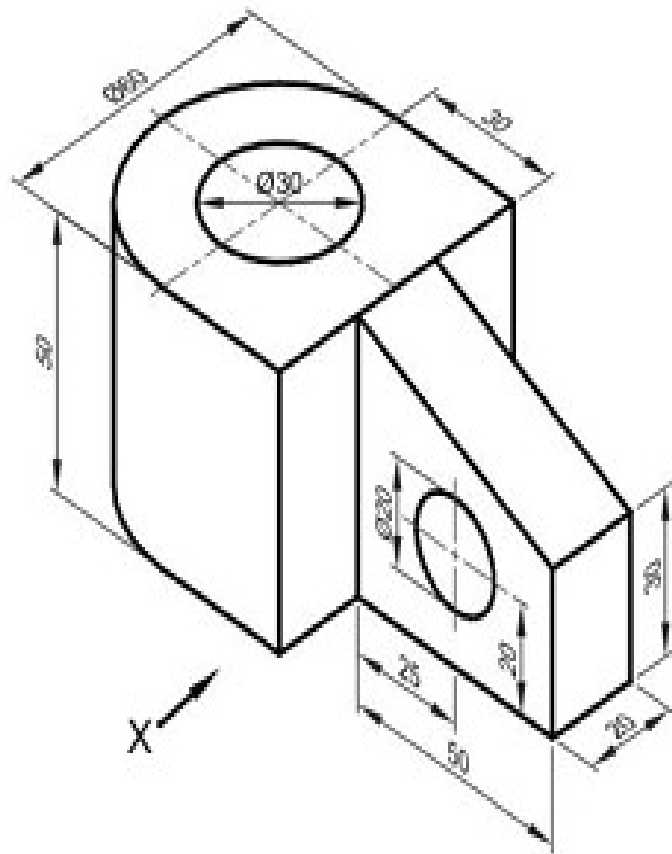
(b)



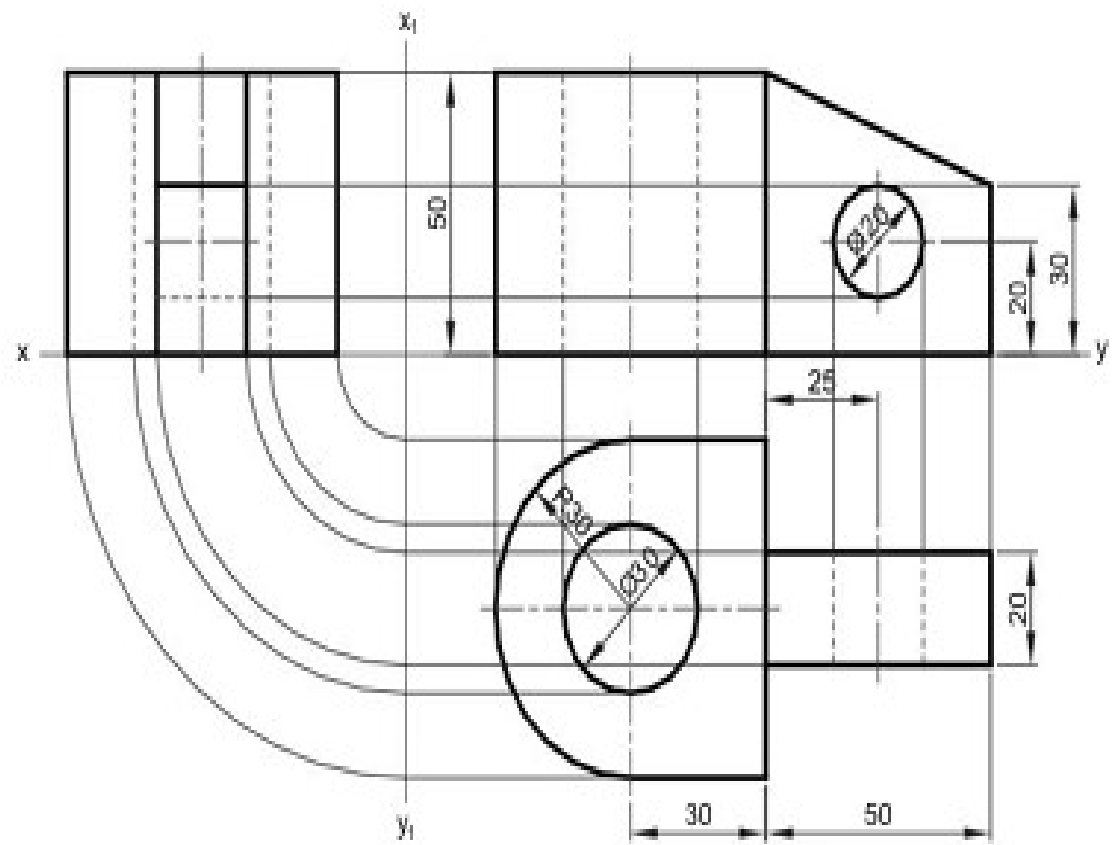
(a)



(b)



(a)

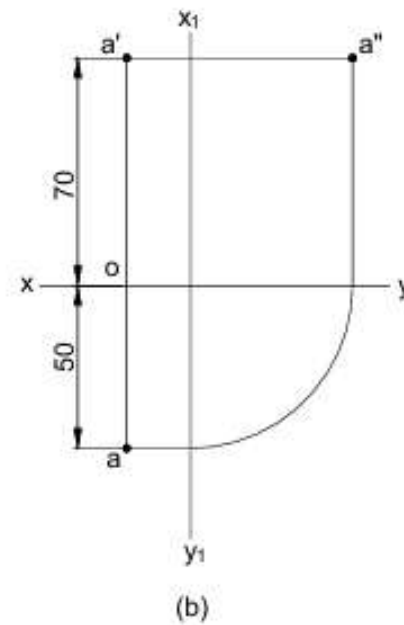
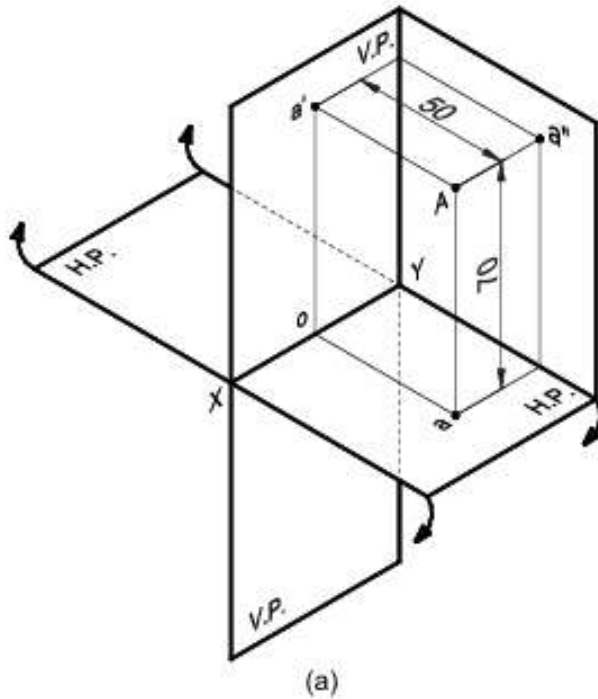


(b)

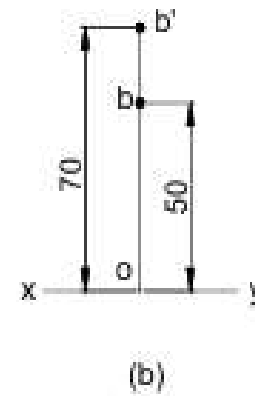
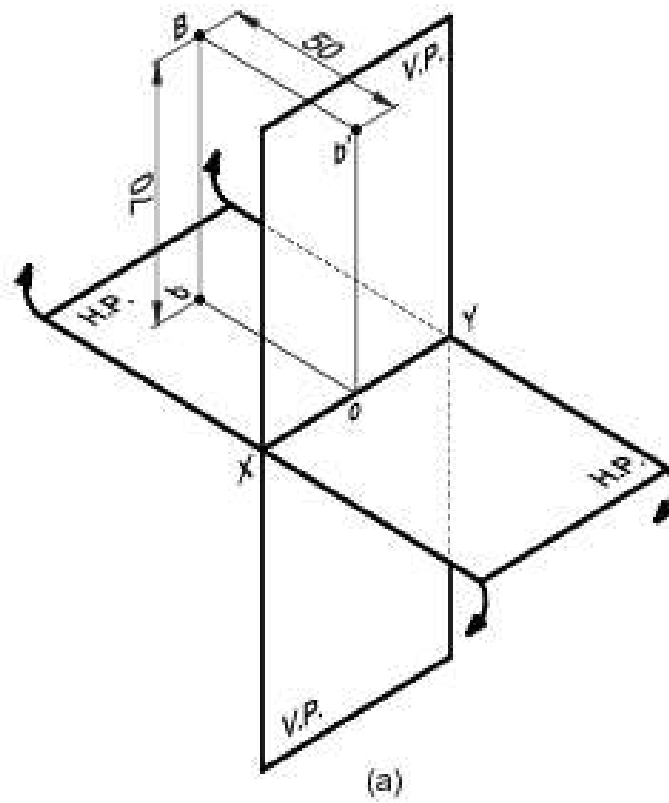
3. PROJECTION OF POINT

A point is defined as a geometrical element that has no dimension. In engineering drawing, a point is represented by a dot.

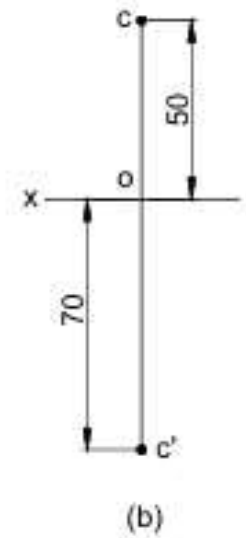
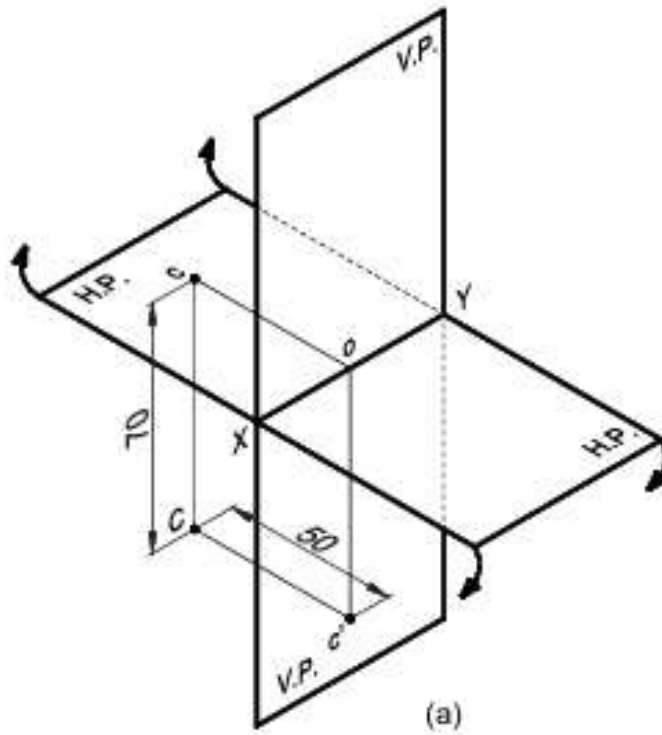
Draw the front and the top views of a point A, lying 70 mm above the H.P. and 50 mm in front of the V.P. Also draw the side view.



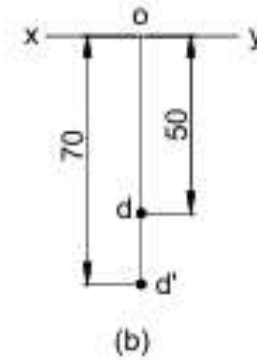
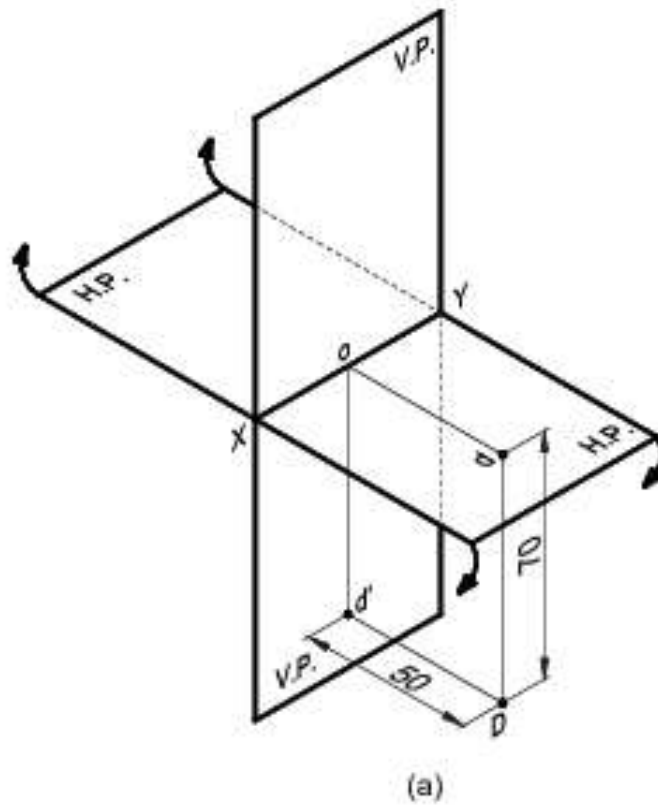
Draw the projections of a point B, lying 70 mm above the H.P. and 50 mm behind the V.P.



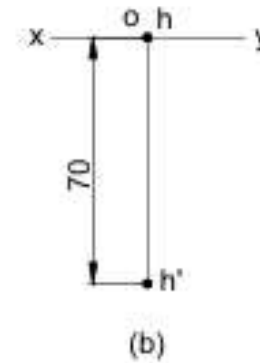
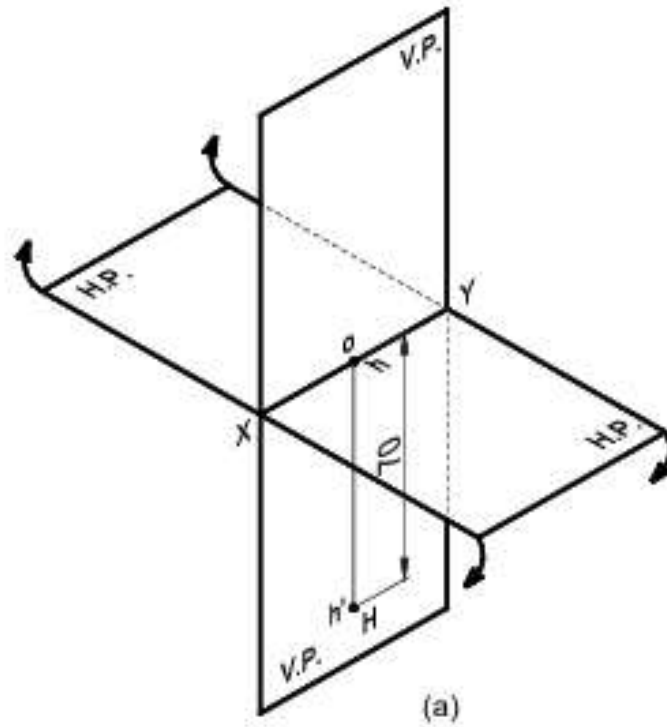
Draw the projections of a point C, lying 70 mm below the H.P. and 50 mm behind the V.P.



Draw the projections of a point D, lying 70 mm below the H.P. and 50 mm in front of the V.P.



Draw the projections of a point H, lying 70 mm above the H.P. and on the V.P



4. Projection of Line

A straight line is defined as the locus of a point which moves unidirectionally. The straight line can also be defined as the shortest distance between two points. The projections of straight lines can be drawn by joining the respective projections of its end points. T

TRACE OF A STRAIGHT LINE

When a line is extended (if needed) and it **meets a reference plane**, that meeting point is called a **trace**.

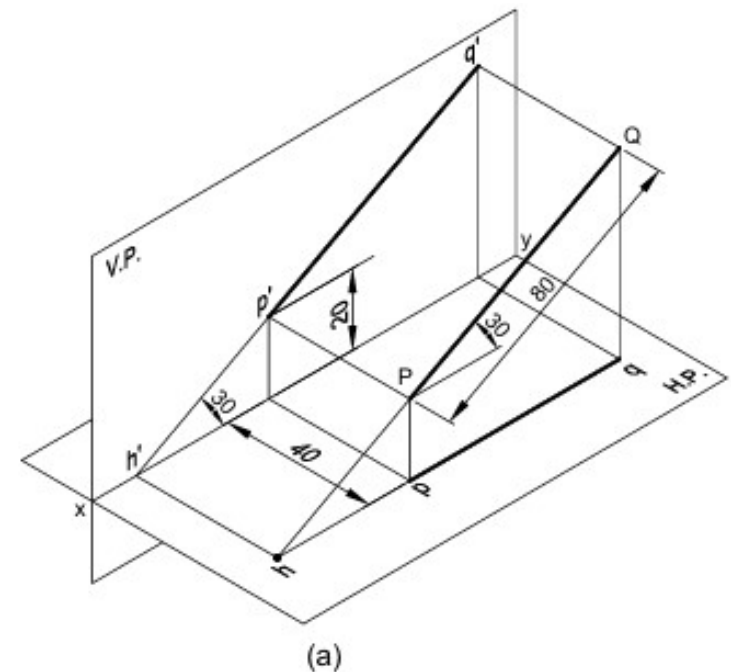
1. Horizontal Trace (H.T.)



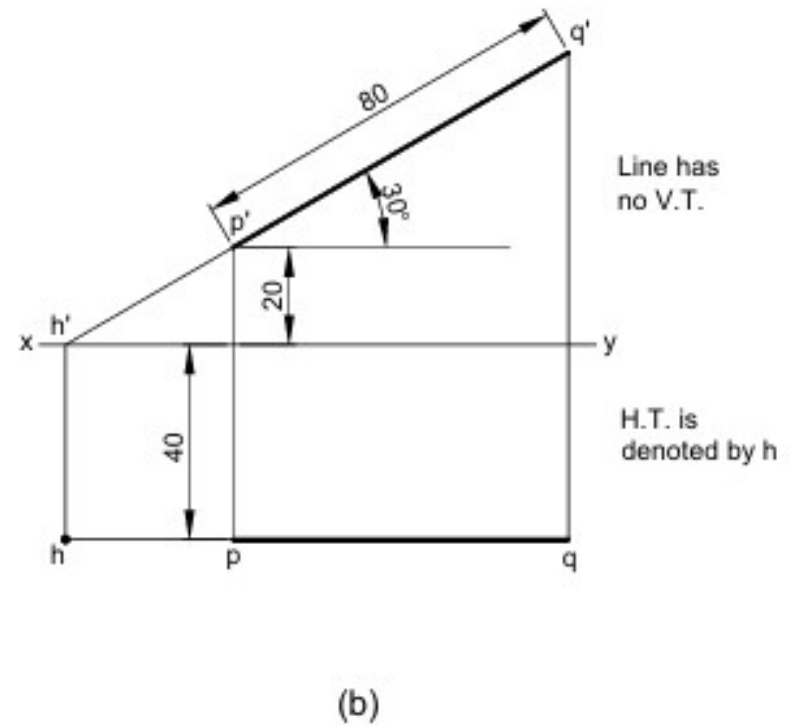
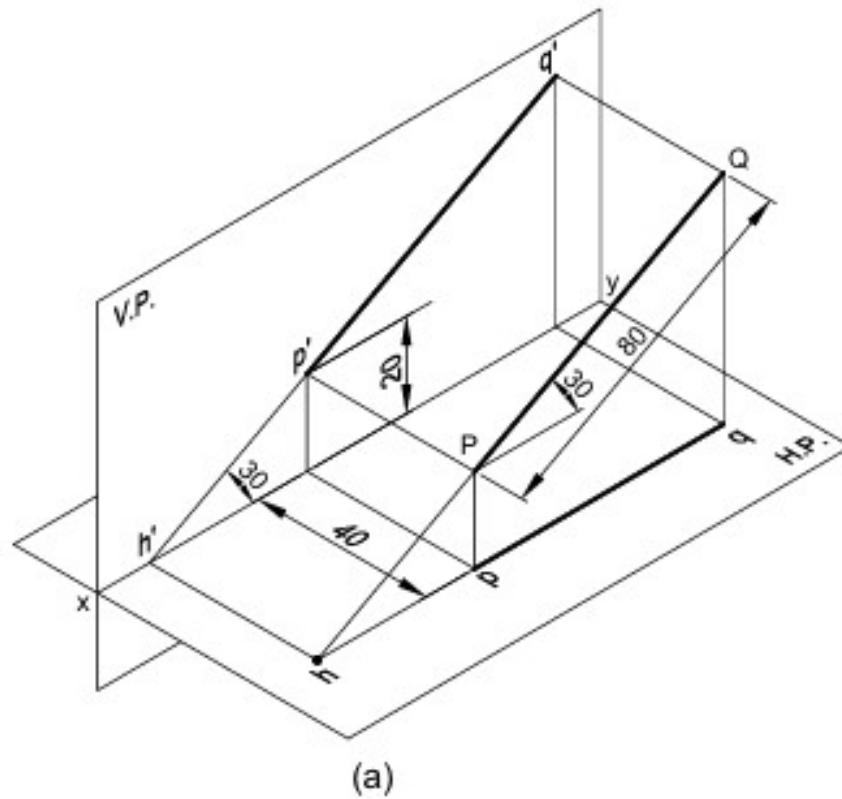
Definition (Simple)

The point where the line **cuts the Horizontal Plane (H.P.)** is called the **Horizontal Trace (H.T.)**.

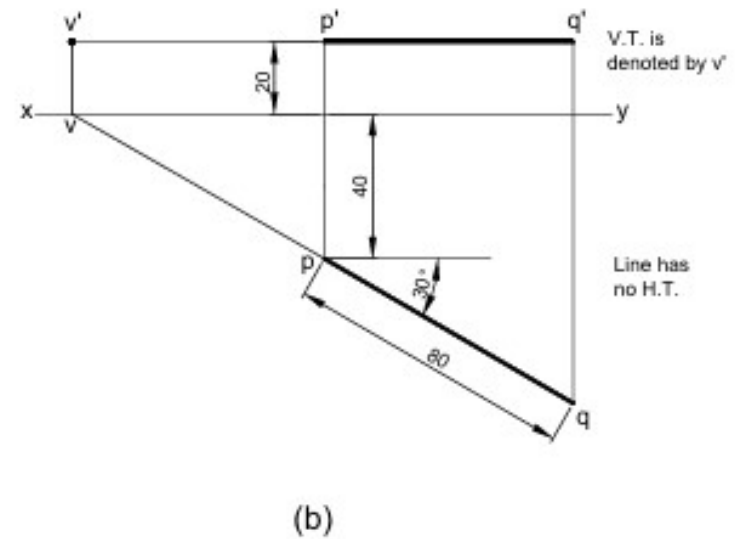
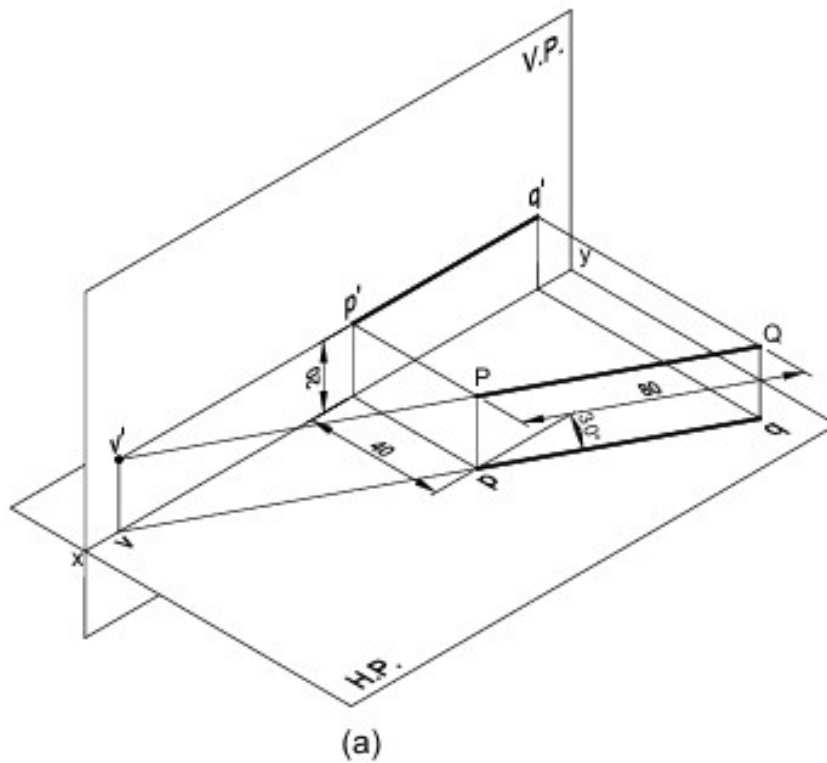
2. The point where the line **cuts the Vertical Plane (V.P.)** is called the **vertical Trace (V.T.)**.



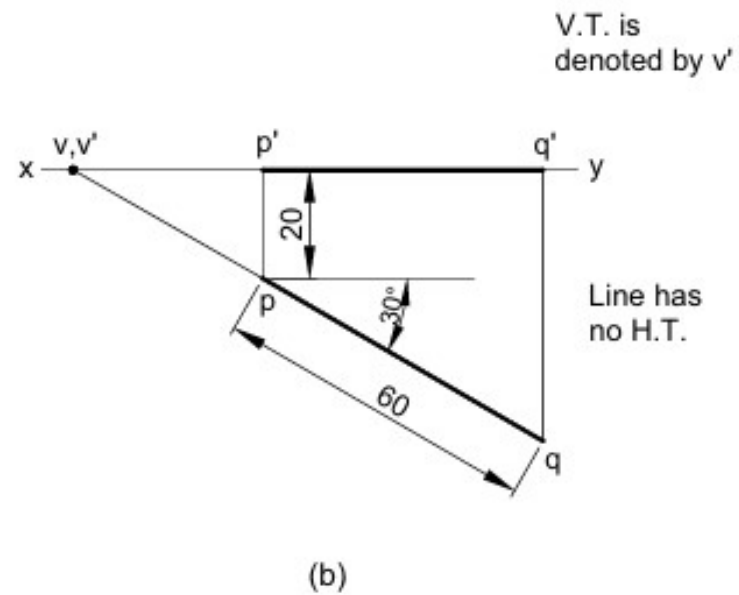
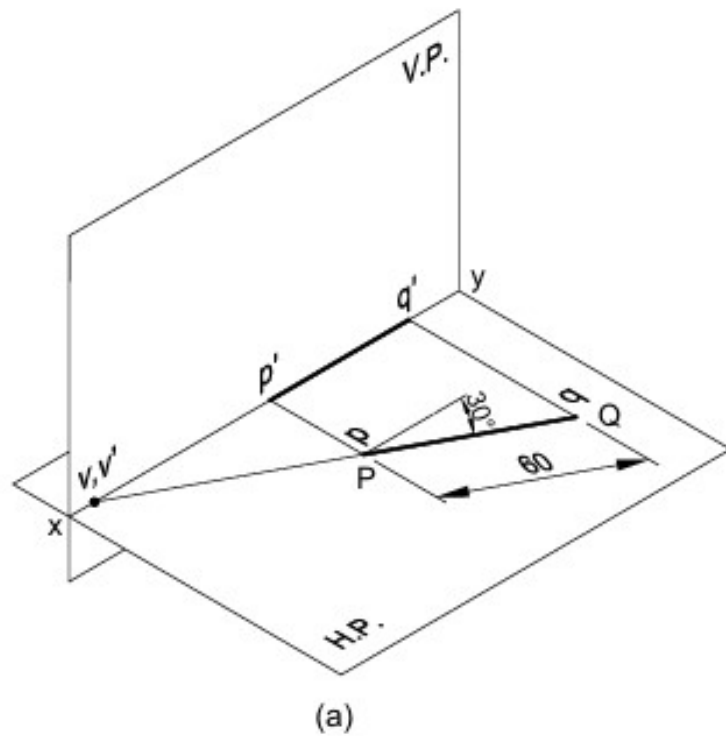
A 80 mm long line PQ has end P 20 mm above H.P. and 40 mm in front of the V.P. The line is inclined at 30° to the H.P. and is parallel to the V.P. Draw the projections of the line and determine its traces



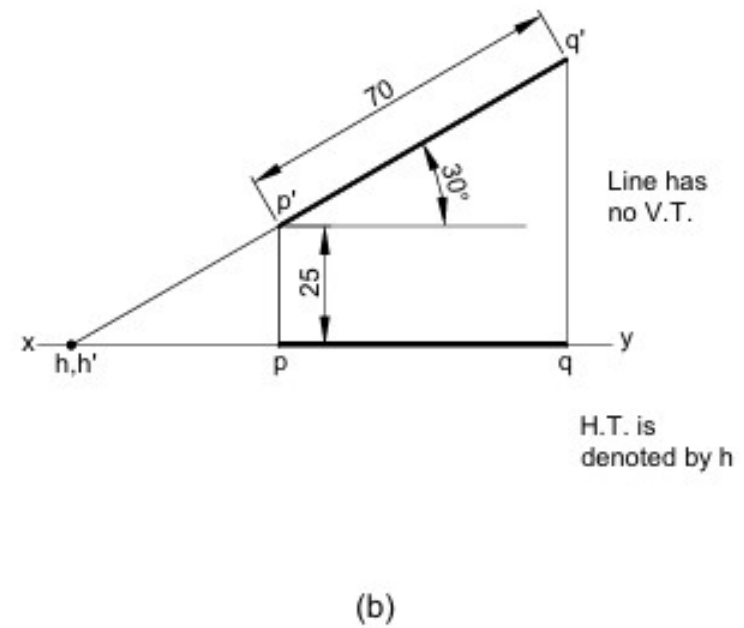
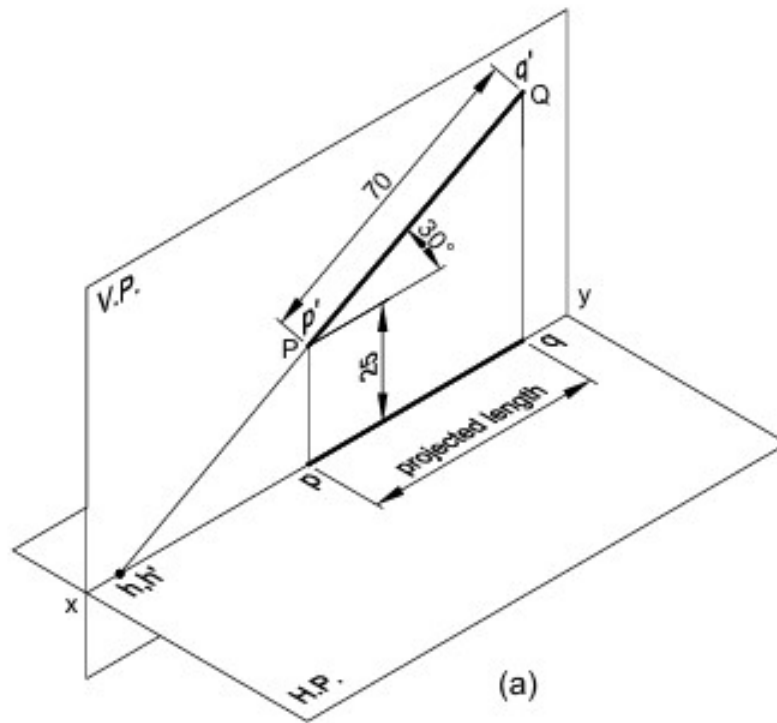
An 80 mm long line PQ is inclined at 30° to the V.P. and is parallel to the H.P. The end P of the line is 20 mm above the H.P. and 40 mm in front of the V.P. Draw the projections of the line and determine its traces



A 60 mm long line PQ lying on the H.P. is inclined at 30° to the V.P. Its end P is 20 mm in front of the V.P. Draw the projections of the line and determine its traces.

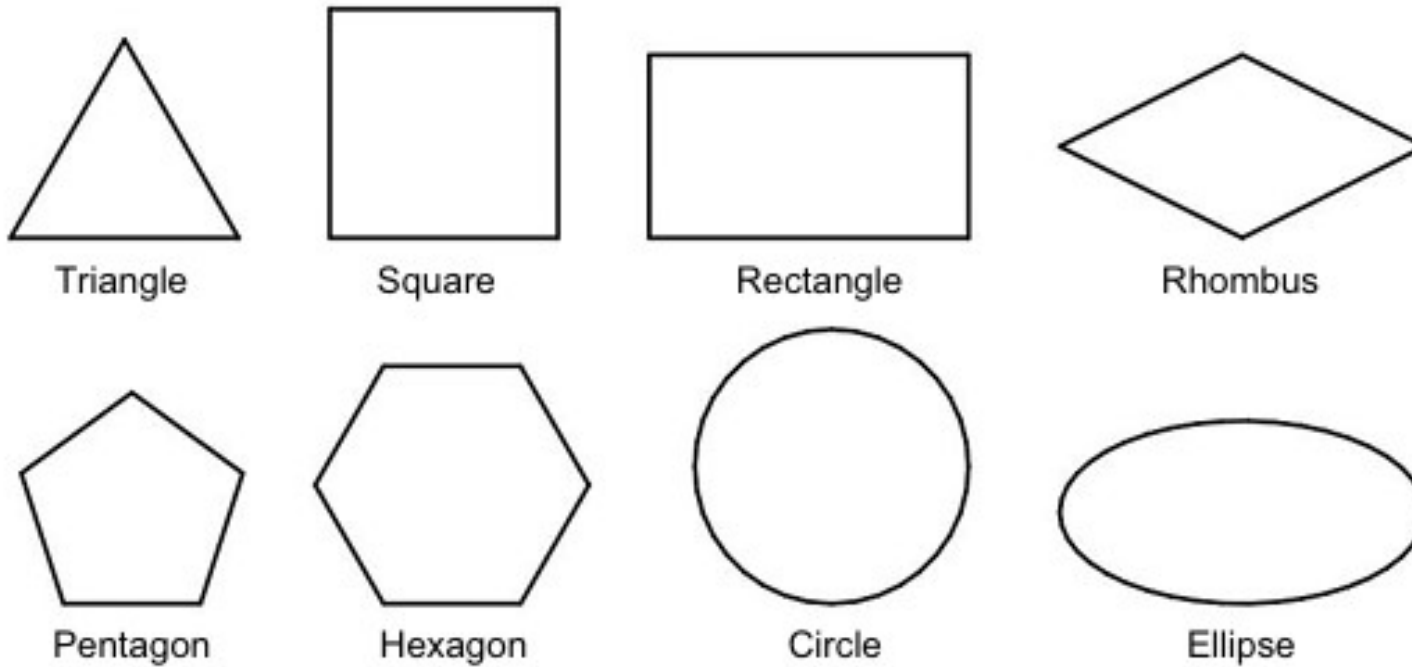


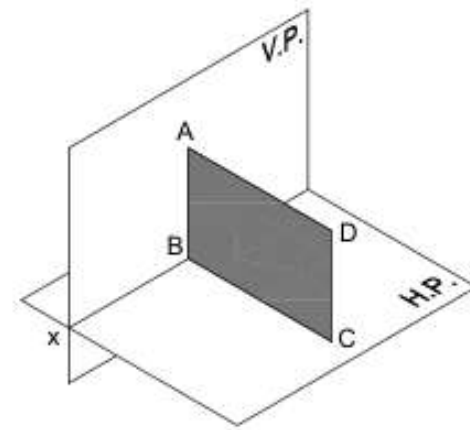
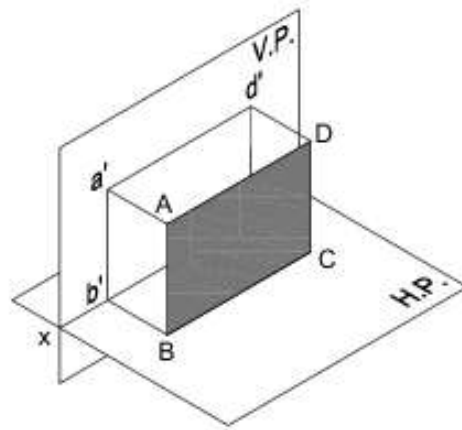
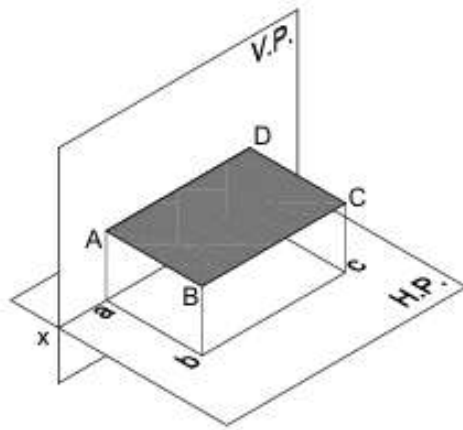
Draw the projections of a 70 mm long line PQ, situated in the V.P. and inclined at 30° to the H.P. The end P of the line is 25 mm above the H.P. Also, determine the traces of the line.



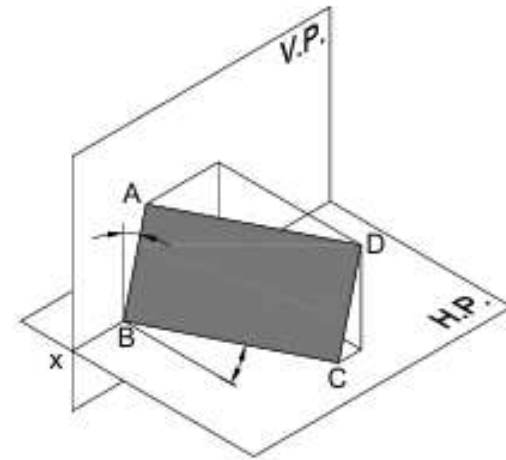
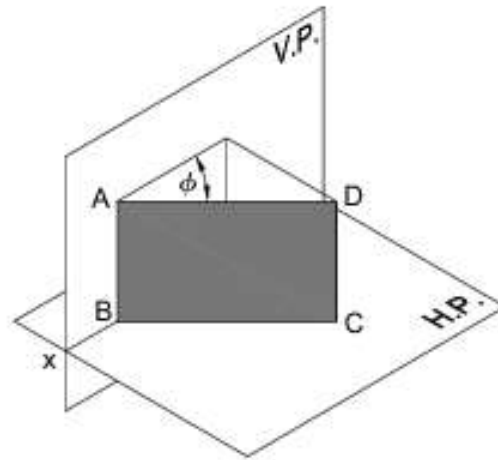
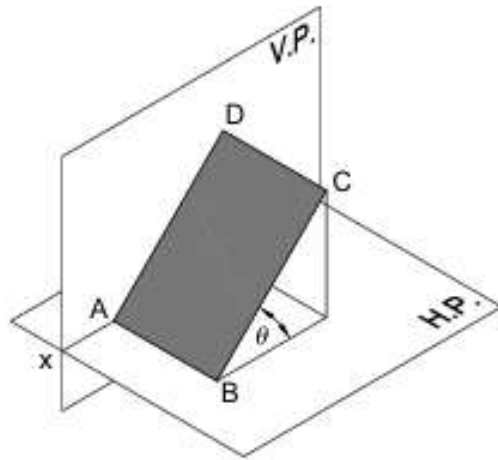
5. PROJECTIONS OF PLANES

we deal with two-dimensional objects called planes. Planes have length, breadth and negligible thickness. Here only those planes are considered whose shape can be defined geometrically and are regular in nature



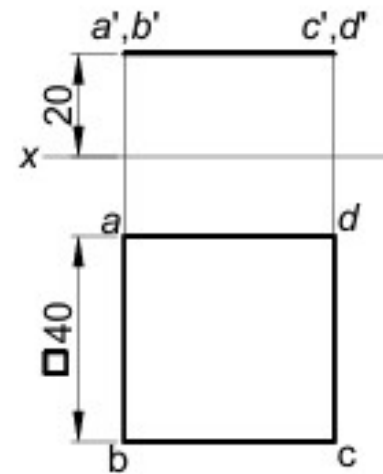
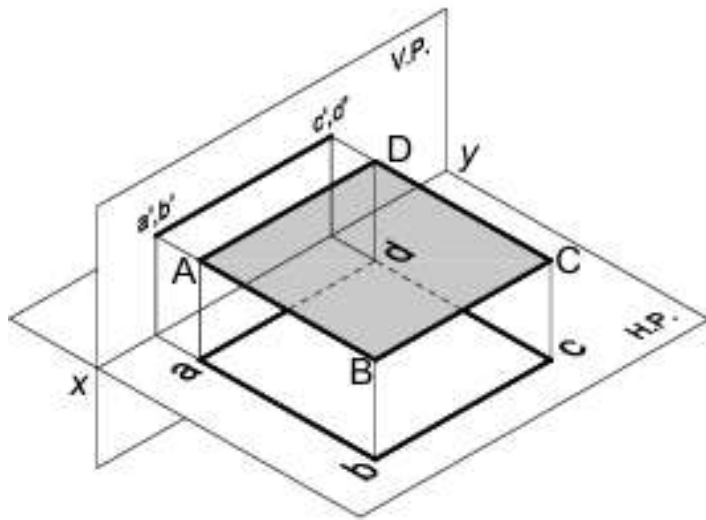


Plane (a) Parallel to H.P. (b) Parallel to V.P. (c) Parallel to profile plane

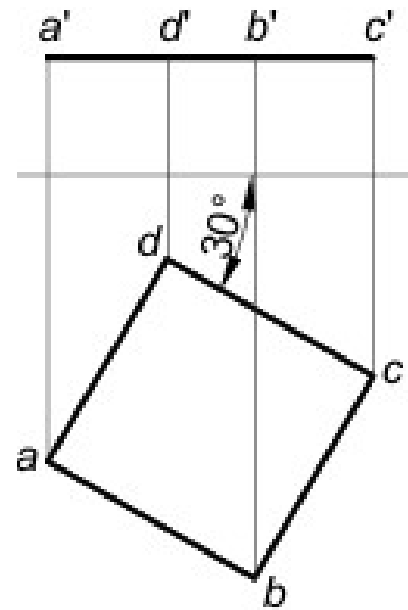
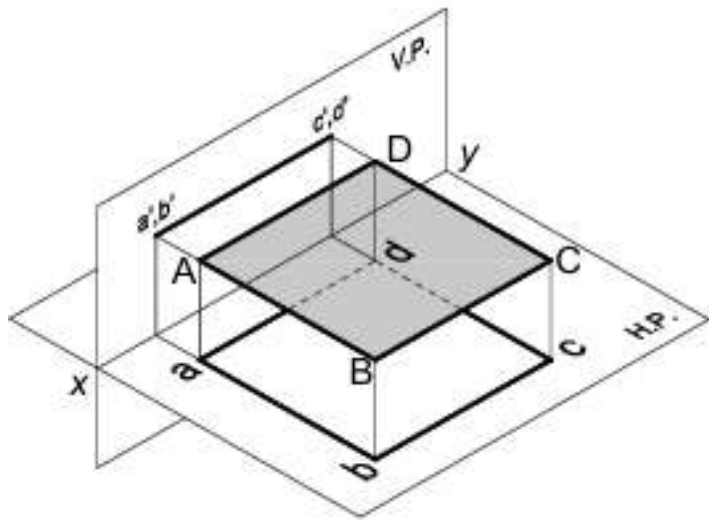


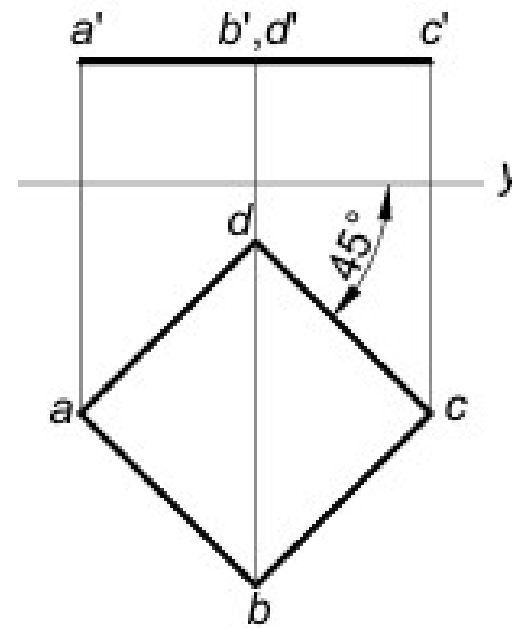
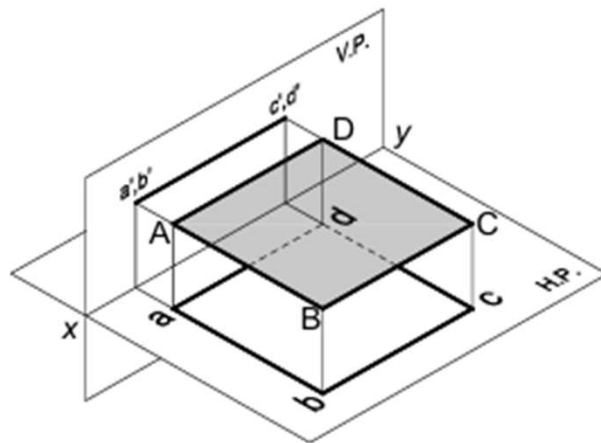
Plane **(a)** Inclined to H.P. and perpendicular to V.P. **(b)** Inclined to V.P. and perpendicular to H.P.
(c) Inclined to both H.P. and V.P.

A square plane of side 40 mm has its surface parallel to and 20 mm above the H.P. Draw its projections when (a) a side is parallel to the V.P., (

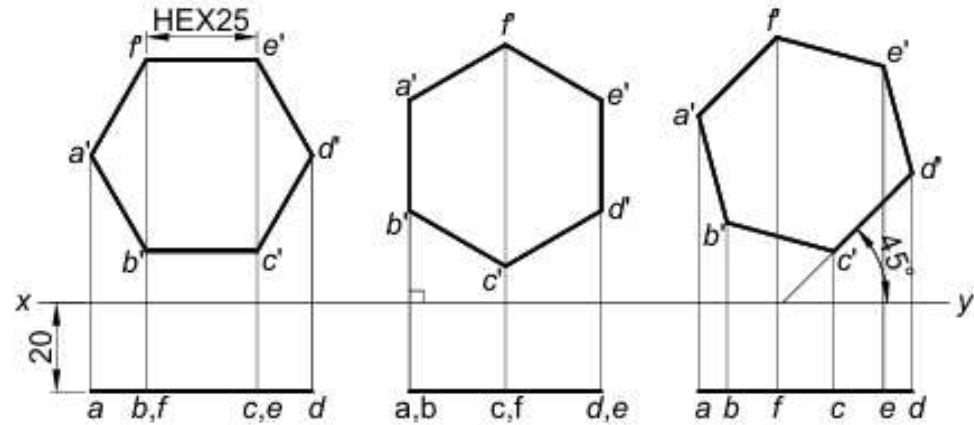
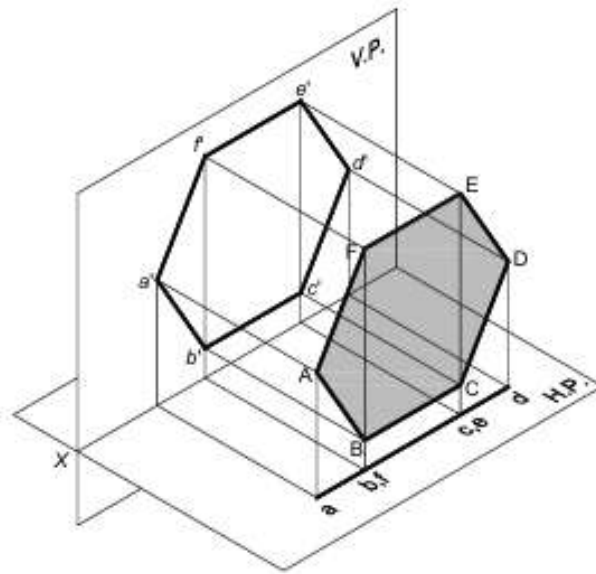


one side is inclined at 30° to the V.P.

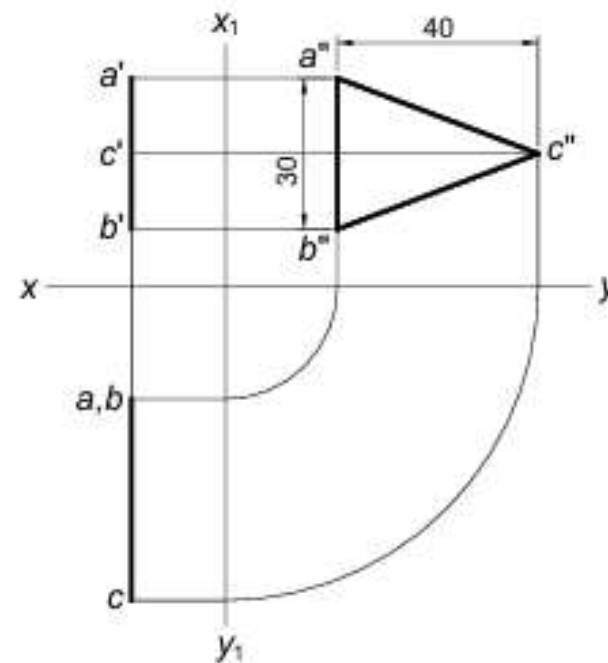
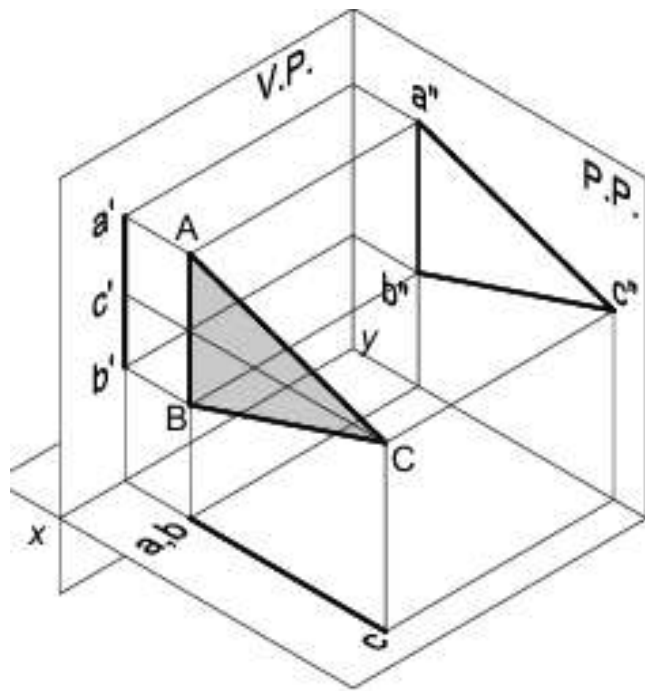




A hexagonal plane of side 25 mm has its surface parallel to and 20 mm in front of V.P. Draw its projections, when a side is (a) parallel to the H.P., (b) perpendicular to the H.P., (c) inclined at 45° to the H.P



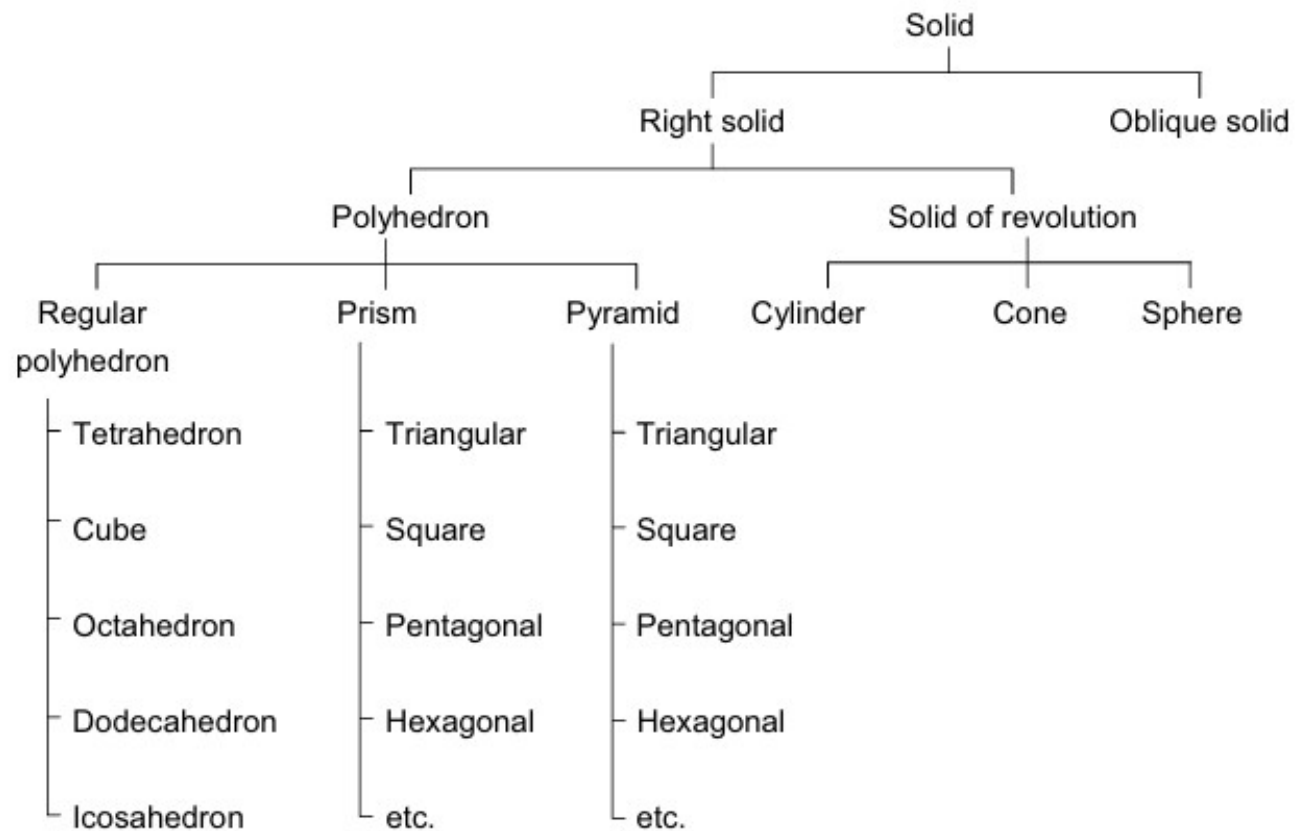
A triangular plane is in the form of an isosceles triangle of base side 30 mm and altitude 40 mm. Its surface is perpendicular to both H.P. and V.P. Draw its projections when the base side is parallel to the V.P.



6. PROJECTIONS OF SOLIDS

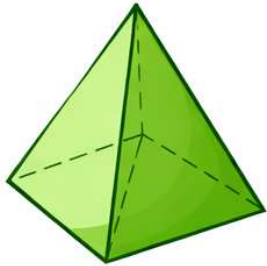
This chapter deals with the orthographic projections of three-dimensional objects called solids. However, only those solids are considered the shape of which can be defined geometrically and are regular in nature.

Solids are usually classified as given below:

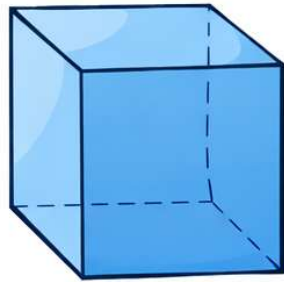


Polyhedron

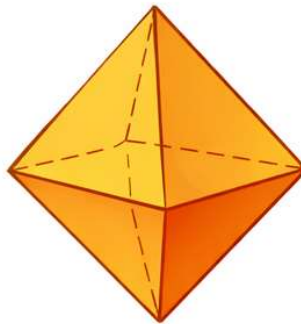
A polyhedron is a solid bounded by planes called faces, which meet in straight lines called edges. A regular polyhedron has all the faces equal and regular.



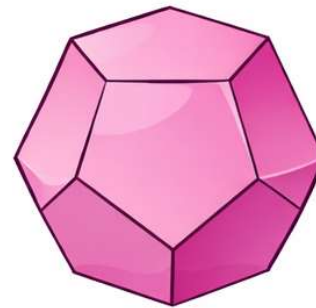
(a)



(b)



(c)



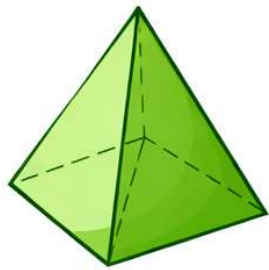
(d)



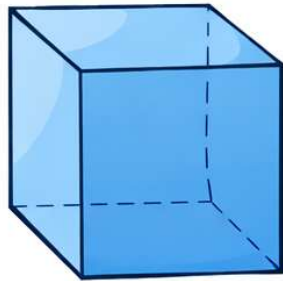
(e)

Regular polyhedron (a) Tetrahedron (b) Cube (c) Octahedron (d) Dodecahedron (e) Icosahedron

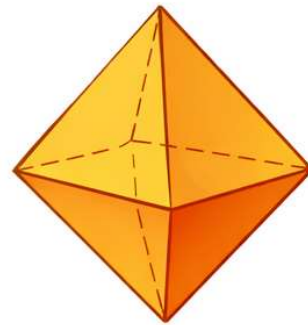
1. **Tetrahedron** It has four equal equilateral triangular faces.
2. **Cube** It has six equal square faces.
3. **Octahedron** It has eight equal equilateral triangular faces.
4. **Dodecahedron** It has 12 equal pentagonal faces.
5. **Icosahedron** It has 20 equal equilateral triangular faces.



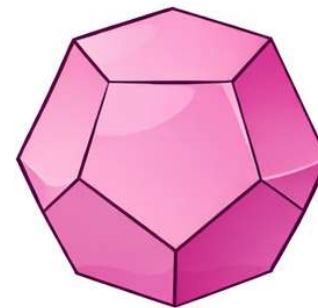
(a)



(b)



(c)



(d)



(e)

Regular polyhedron (a) Tetrahedron (b) Cube (c) Octahedron (d) Dodecahedron (e) Icosahedron

A **prism** is a 3D shape that has:

- **Two same (equal) faces** on top and bottom (called bases)
- These bases are **parallel**
- The side faces are **rectangles**

Right & Regular Prism (Very Important)

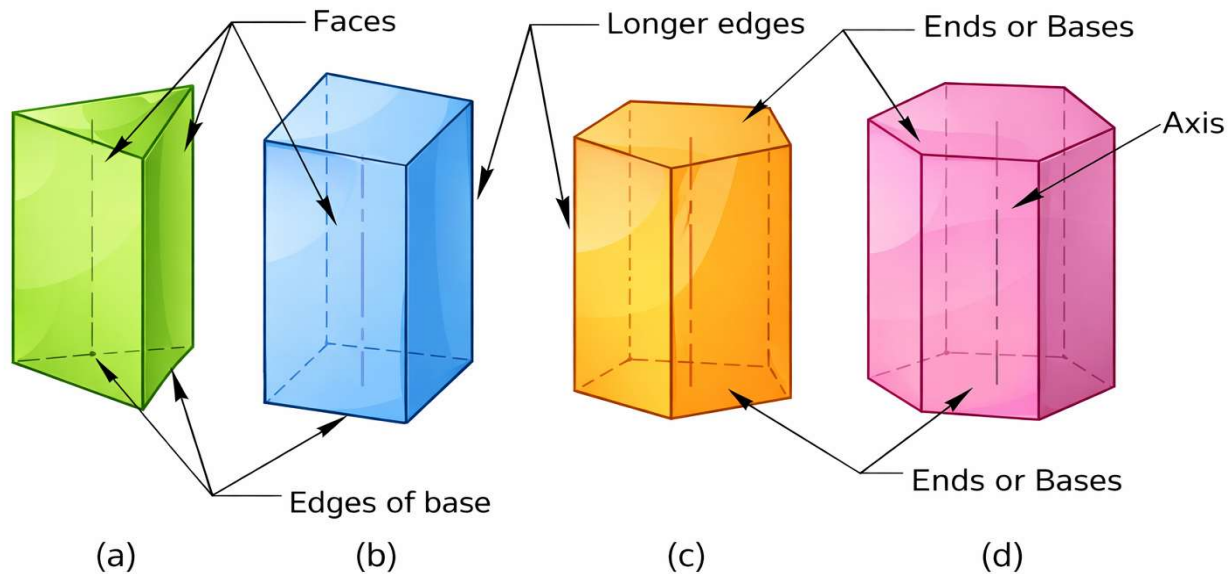
A prism is called **Right & Regular** when:

- Base is a **regular shape** (all sides equal)
- Axis is **perpendicular (90°)** to the base
- All side faces are **equal rectangles**

Feature	Meaning (Simple)
Bases	Top and bottom faces (same shape & size)
Parallel	Bases never meet (always same distance)
Lateral Faces	Side faces → always rectangles
Axis	Imaginary line joining center of top & bottom
Cross-section	Any cut parallel to base looks same as base

Base Shape	Name of Prism
Triangle	Triangular Prism
Square	Square Prism
Pentagon	Pentagonal Prism
Hexagon	Hexagonal Prism

Feature	Meaning (Simple)	Base Shape	Name of Prism
Bases	Top and bottom faces (same shape & size)	Triangle	Triangular Prism
Parallel	Bases never meet (always same distance)	Square	Square Prism
Lateral Faces	Side faces → always rectangles	Pentagon	Pentagonal Prism
Axis	Imaginary line joining center of top & bottom	Hexagon	Hexagonal Prism
Cross-section	Any cut parallel to base looks same as base		



Pyramid

A **pyramid** is a 3D shape that has:

- **One base** (any polygon: triangle, square, etc.)
- **Side faces are triangles**
- All triangles meet at **one point (top)** → called **apex**

A pyramid is **Right & Regular** when:

- Base is a **regular shape** (all sides equal)
- Axis is **perpendicular (90°)** to base
- All side faces are **equal isosceles triangles**

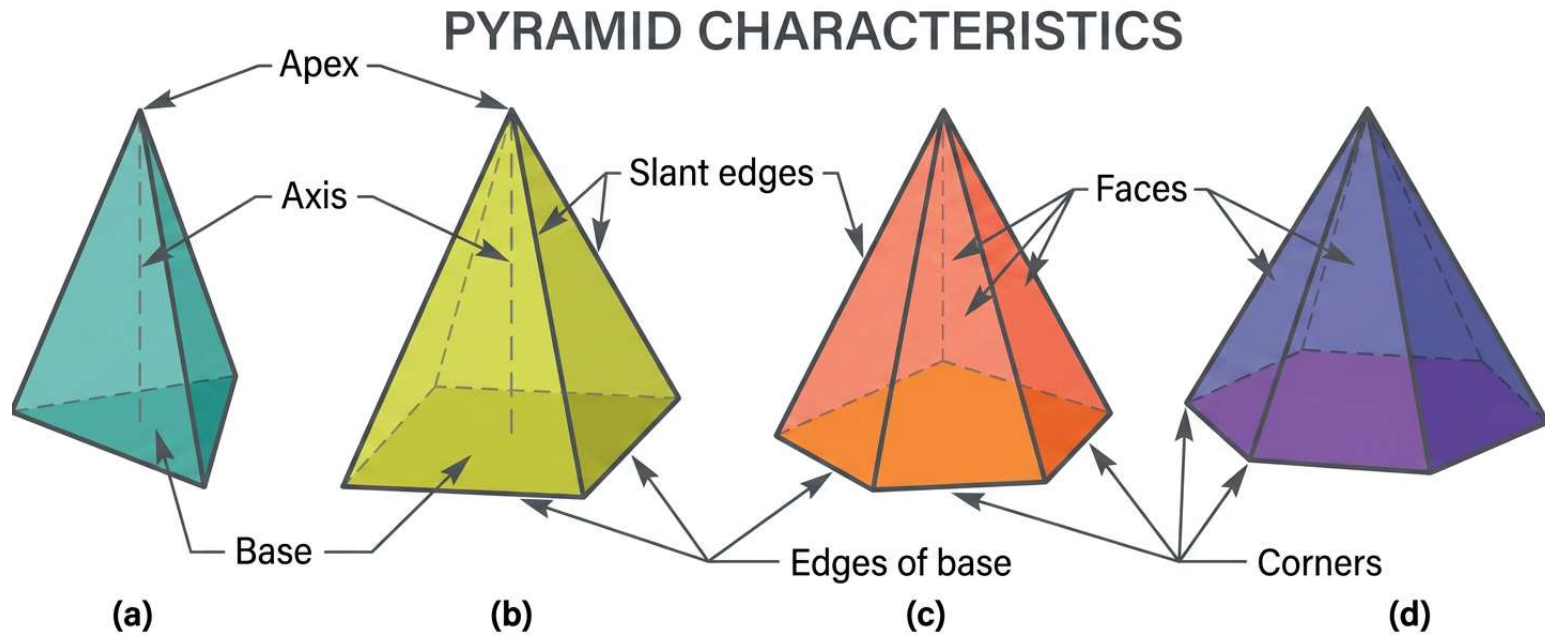
Feature	Meaning (Very Simple)
Base	Bottom shape (triangle, square, etc.)
Lateral Faces	Triangular sides
Apex (Vertex)	Top point where all triangles meet
Axis	Line from apex to center of base
Name	Depends on base shape

The **centre of gravity (CG)** lies:

- On the **axis**
- At **1/4th of height from the base**

Base Shape	Name
Triangle	Triangular Pyramid
Square	Square Pyramid
Pentagon	Pentagonal Pyramid
Hexagon	Hexagonal Pyramid

Feature	Meaning (Very Simple)	Base Shape	Name
Base	Bottom shape (triangle, square, etc.)	Triangle	Triangular Pyramid
Lateral Faces	Triangular sides	Square	Square Pyramid
Apex (Vertex)	Top point where all triangles meet	Pentagon	Pentagonal Pyramid
Axis	Line from apex to center of base	Hexagon	Hexagonal Pyramid
Name	Depends on base shape		



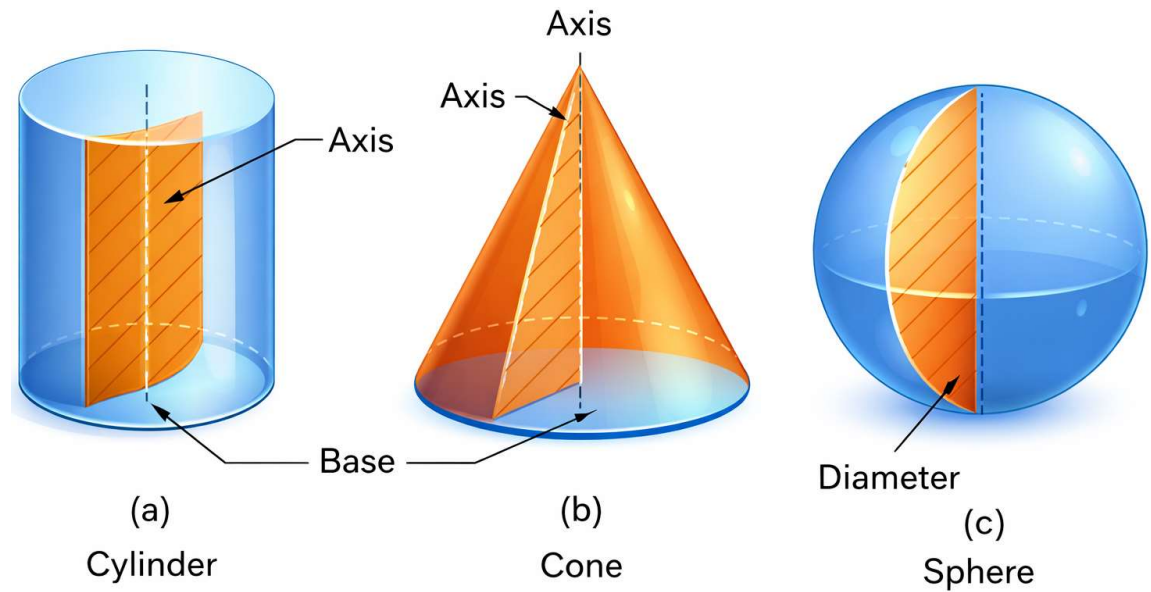
4 Pyramids (a) Triangular (b) Square (c) Pentagonal (d) Hexagonal

Solid of Revolution

- Imagine a **rectangle**
Now rotate it around one of its sides (like spinning a door)
- What do you get? → **Cylinder (like a pipe or glass)**

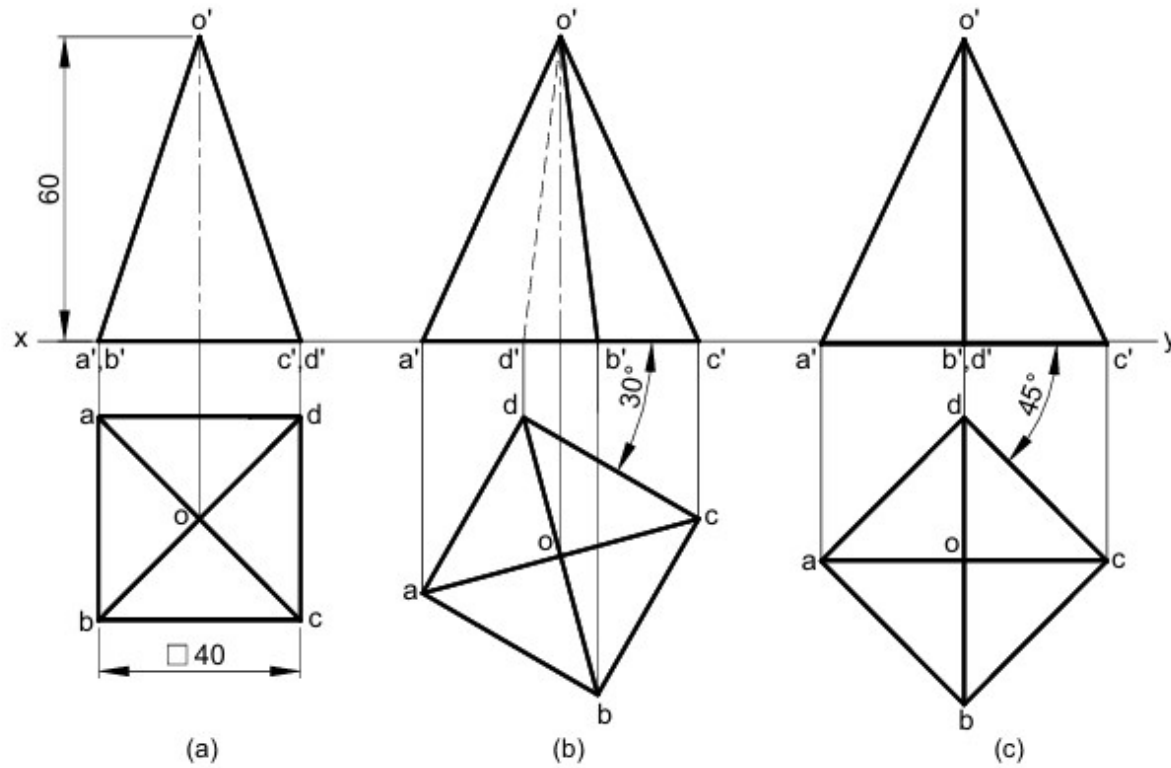
- Imagine a **triangle**
Rotate it around one of its sides
- What do you get? → **Cone (like an ice-cream cone)**

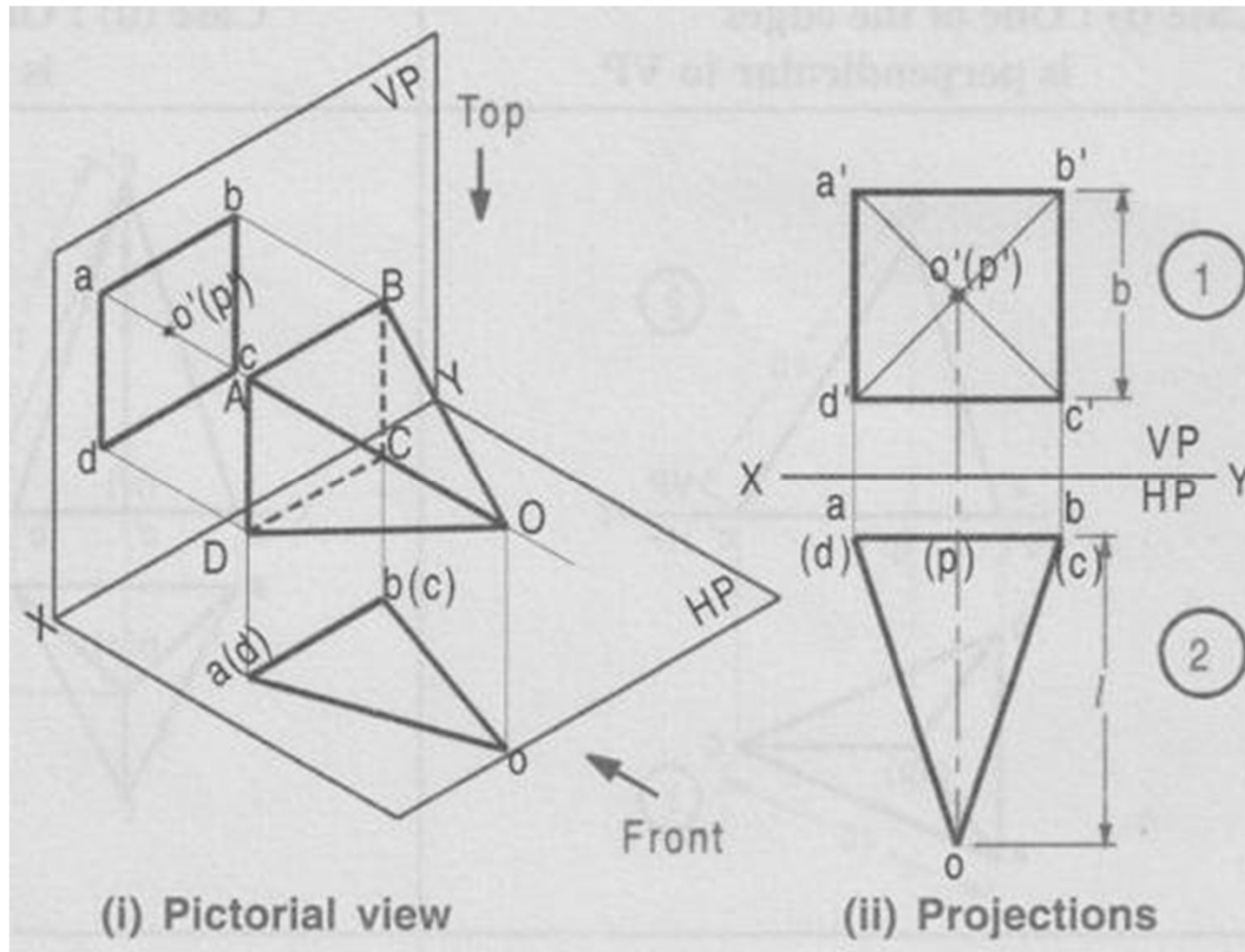
- Imagine a **half circle (semicircle)**
Rotate it around its diameter
- What do you get? → **Sphere (like a ball)**



A square pyramid of base side 40 mm and axis 60 mm is resting on its base on the H.P.

Draw its projections when (a) a side of the base is parallel to the V.P., (b) a side of the base is inclined at 30° to the V.P., (c) all the sides of the base are equally inclined to the V.P.





A hexagonal prism of base edge 30 mm and axis 70 mm has its axis parallel to and 50 mm above the H.P. Its base is parallel to the V.P. and an edge of the base is inclined at 45° to the H.P. Draw its projections.

